

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.	ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
58			6-32 x 3/8" BINDER HEAD SCREW	3	48	5A-6314	F1, F2	4 AMP SLOW BLOW FUSE	2
59			6-32 HEX NUT	3	49	5A-9178		FUSEHOLDER	4
60		W1, W2, W6, W7, W9, W10, W15	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	7	50	5A-9172		HEAT SINK THERMALLOY #6072B	1
61	5A-9248	TP1 THR TPA	TERMINAL #1502-1	4	51	5A-9173		HEAT SINK THERMALLOY #6071B	1
62	5A-9363	R11	RESISTOR, FC, 5.6 K OHM 5% 1/4 WATT	1	52	5A-9199		HEAT SINK THERMALLOY #6030	1
64	5A-9362	SRI	RESISTOR, 4.7 K OHM 10 PIN SIP	1	53	5A-9004		24 PIN SOCKET	1
65	3A-7520-1		TIE WRAP	1	54	5A-B985		40 PIN SOCKET	1
					55	5A-9027	10J1, 10J3	9 PIN MALE CONNECTOR	2
					56	5A-9028	10J2, 10J4	4 PIN MALE CONNECTOR	2
					57	5A-9349	10J5	40 PIN RIBBON HEADER	1

BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	1C-2001-146-3		BARE P.C. BOARD	1
2	5A-9156	IC1	TDA2002 V AUDIO AMPLIFIER	1
3	5A-9012	IC2	7442 BCD-DEC DECODER	1
4	5A-9073	IC3	7400 QUAD 2 INPUT NAND	1
5	5A-8973	IC4	7408 QUAD 2 INP. AND GATE	1
6	5A-9153	IC5	4050 BUFFER	1
7	5A-9154	IC6	4068 BINPUT NAND GATE	1
8	5A-B971	IC7	14069 HEX INVERTER	1
9	5A-9157	IC8	7805 5 VOLT REG. W/TO220 CASE	1
10	5A-B972	IC10	6821 P.I.A.	1
11	5A-9003	IC11	6810 RAM	1
12	5A-9152	IC13	1408 D/A CONVERTER	1
13	5C-B938	Q2, Q3, Q4	2N4401 NPN TRANSISTOR	3
14				
15	5A-9018	ZR1	1N5996 6.8V. ZENER DIODE	1
16				
17	5A-9158 OR 5A-9357 5A-9020	BR1	MDA 200/3N253 BRIDGE RECTIFIER	1
18		Y1	3.58 MHz CRYSTAL	1
19	5B-B991	R1, R18, R19, R21, R22, R27, R30, R31, R32	RESISTOR, FC, 4.7K OHM 5% 1/4 WATT	9
20	5B-9036	R2 THRU R10	RESISTOR, FC, 100 OHM 10% 1/4 WATT	9
21	5A-B984	R12, R15, R28, R36, R38	RESISTOR, FC, 1K OHM 10% 1/4 WATT	5
22	5A-9181	R14	RESISTOR, FC, 1 OHM 10% 1/2 WATT	1
23	5A-9161	R16	RESISTOR, FC, 2.2 OHM 10% 1/4 WATT	1
24	5A-9361	R17	RESISTOR, FC, 220 OHM 10% 1/2 WATT	1
25				
26	5B-B988	R23, R24, R26	RESISTOR, FC, 3.3K OHM 10% 1/4 WATT	3
27	5A-9179	R25	RESISTOR, FC, 3.3M OHM 10% 1/4 WATT	1
28	5A-9359	R29	RESISTOR, FC, 47 K OHM 5% 1/4 WATT	1
29	5B-B817	R33, R35, R37	RESISTOR, FC, 10 K OHM 10% 1/4 WATT	3
30	5B-9039	R34	RESISTOR, FC, 10 OHM 10% 1/4 WATT	1
31	5A-B980	C1, C16 THRU C23	CAPACITOR, CERAMIC, .01 MFD. 50 V. ±20%	11
32	5A-9065	C2 THRU C10	CAPACITOR, CERAMIC, 470 PFD. 50 V. ±20%	9
33	5A-9345	C11	CAPACITOR, CERAMIC, .001 MFD. 20% 100 V.	1
34	5A-9365	C12, C30, C36	CAPACITOR, ELECTROLYTIC 1 MFD. 63 V. 10% ±20%	3
35	5A-B996	C13, C24, C35	CAPACITOR, CERAMIC, .1 MFD. 50 V. ±20%	3
36	5A-9165 5A-9165-1	C14	CAPACITOR, ELECTROLYTIC, 800 MFD. 16 V. OR 1,000 MFD. 15 V. ±20%	1
37	5A-9164 5A-9164-1	C15	CAPACITOR, ELECTROLYTIC, 500 MFD. 15 V. OR 470 MFD. 25 V. ±20%	1
38	5A-B986	C25	CAPACITOR, ELECTROLYTIC, 100 MFD. 10 V. ±20%	1
39	5A-B893	C26	CAPACITOR, ELECTROLYTIC, 1,000 MFD. 25 V. ±20%	1
40	5A-9046	C27	CAPACITOR, ELECTROLYTIC, 12,000 MFD. 16 V. ±20%	1
41	5A-9180	C28	CAPACITOR, CERAMIC, 47 PFD. 1K V. ±20%	1
42	5A-9343	C29	CAPACITOR, ELECTROLYTIC, 10 MFD. 25 V. LOW LEAK 22%	1
43	5A-9169	C32, C33	CAPACITOR, CERAMIC DISC, 27 PFD. 1K V. ±10%	2
44	5A-9168	C34	CAPACITOR, TANTALUM, 2.2 MFD. 15 V. ±20%	1
45	5A-9031	C37	CAPACITOR, TANTALUM, 1 MFD. 25 V. ±20%	1
46	5A-9024	SW1	MOMENTARY SWITCH SPDT	1
47	5A-9380	DS1	2 STD. DIP SWITCH	1

NOTES:

- USE THERMAL COMPOUND BETWEEN IC8 AND HEAT SINK.
- CAUTION: AVOID STATIC DISCHARGE DAMAGE TO MOS LOGIC.
- SYMBOLS SHOWN ON COMPONENTS ARE FOR REFERENCE ONLY. DO NOT SCREEN OR STAMP.
- OBSERVE INDEX MARK OF ALL INTEGRATED CIRCUITS, DIODES D1, D2, AND ZR1.
- CAPACITORS C12, C14, C15, C25, C26, C27.
- CONNECTORS 10J1, 10J2, 10J4, 10J3, 10J5.
- POSITION OF TRANSISTORS Q1, Q2, Q3, Q4.

- DS1 - 1 SELECTS SOUND/NOTES
- SELECTS SPEECH/NO SPEECH (W9/W4)
- W1 - SPEECH MODULE STATUS
- IN - SPEECH MODULE NOT ATTACHED
- OUT - SPEECH MODULE ATTACHED
- W14 - MPU INTERNAL RAM ENABLE
- W7 & W8 - MEMORY MAP CONTROL
- W12 & W13 - P07 STATUS CONTROL (W13 NEVER USED)
- W4 & W9 - P05 STATUS CONTROL
- SOLDERED ON TOP OF BOARD
- INSTALL THESE JUMPERS FOR FOLLOWING GAMES:
- W15, W6, W12, W4, W1, W3, W6, W11 FOR:

WORLD CUP
DISCO FEVER
CONTACT
POKIRINO
PHOENIX
ARISTOCRAT SHUFFLE
POMPEII SHUFFLE
KING TUT SHUFFLE
TAURUS SHUFFLE

SOUND ROM 1

W15, W6, W12, W4, W1, W3, W6, W11 FOR:

FLASH
STELLAR WARS
TRI ZONL
TIME WARP

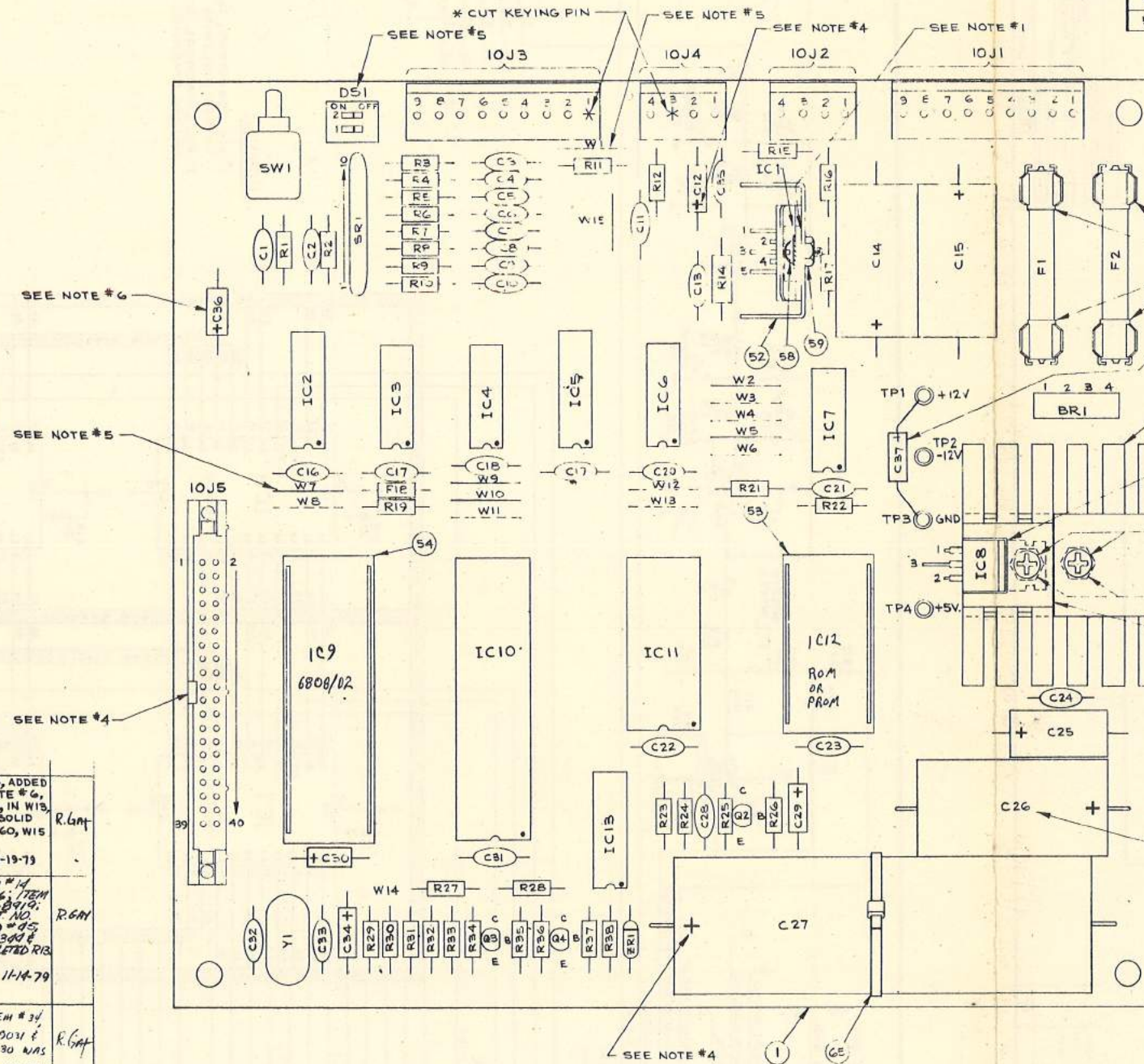
SOUND ROM 1

W7, W15, W9, W1, (SEE NOTE #5) W2, W5, W10 FOR:

GORGAR; SOUND ROM 2 - JT-4960

IC 12 SELECTION STRAPPING:

(2K x 8)	(4K x 8)	(512 x 8)
W2	W3	W3
W5 IN	W3 IN	W6 IN
A10	A10	W11



E	REVISED NOTES, ADDED ITEM #45 & NOTE #6, RELOCATED W15, IN W13, DASH LINE WAS SOLID LINE & ITEM #60, W15 WAS W13 E.C.O.	11-19-79	R.G.M.
D	DELETED ITEM #14, PT. NO. 5A-9016, ITEM #16, PT. NO. 5A-8919, ITEM #25, PT. NO. 5A-9314, ITEM #45, PT. NO. 5A-9360 & IN ITEM #21, DELETED R13, QTY. WAS 6 E.C.O. 4764	11-14-79	R.G.M.
C	ADDED C36, ITEM #34, PT. NO. WAS 5A-9031 & QTY. WAS 1, & C30 WAS 01 MFD. E.C.O. 9760	10-29-79	R.G.M.
B	DELETED ITEM #62, ADDED W1 & R12 TO ITEM #14, QTY. WAS 1, W1 WAS 1, R12 WAS 1	10-27-79	R.G.M.
A	ITEM #11, A10, 5A-9511, ITEM #52, A10, 5A-9511, ITEM #61, A10, 5A-9511	10-4-79	R.G.M.

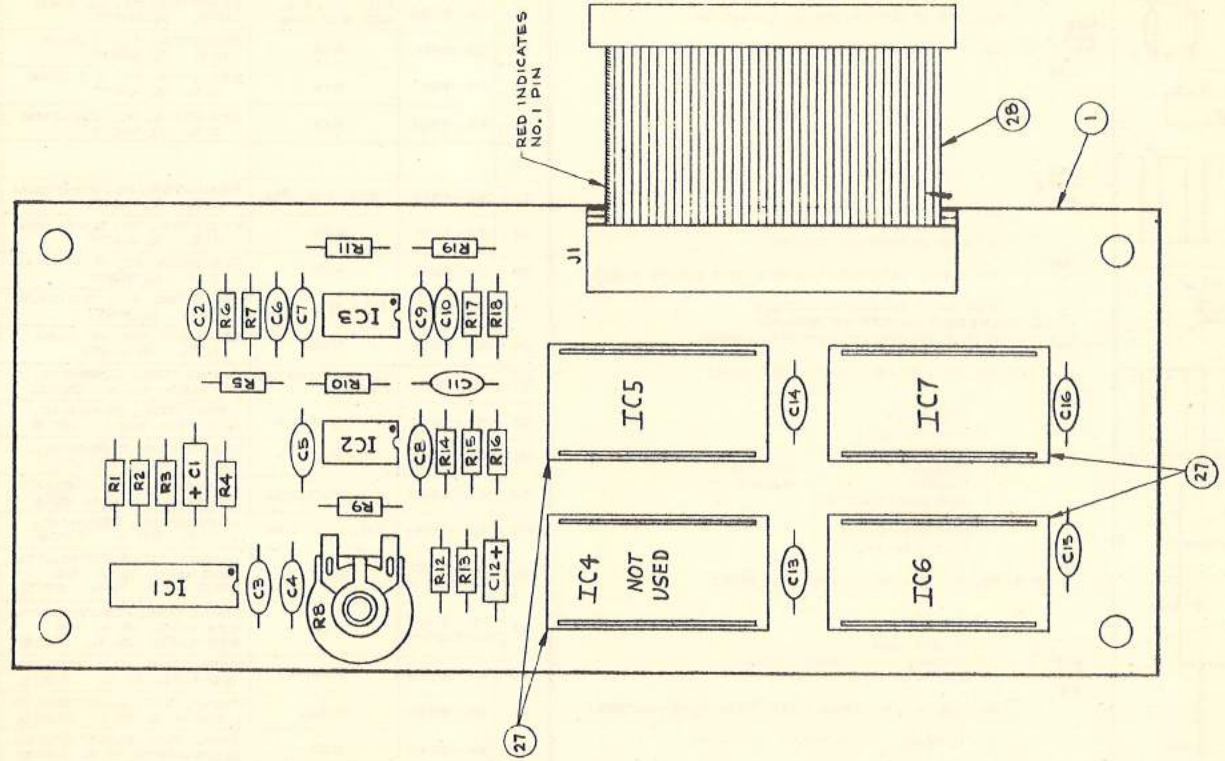
REVISION	DATE	BY
1	10-4-79	R.G.M.

D8224EN NO. 188H AGEPROOF

FIG. 1000-1000-1000

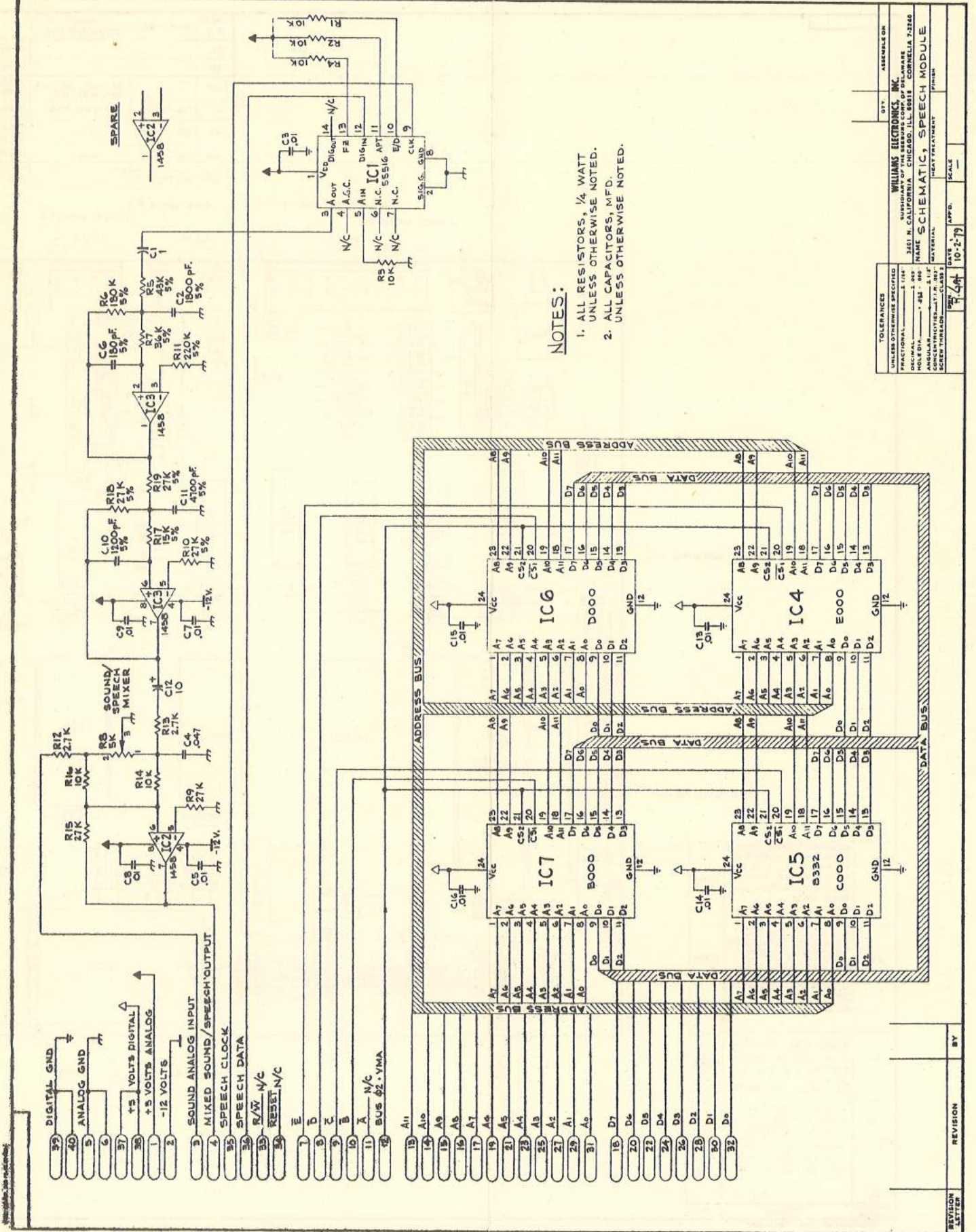
BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	RECD. NO.
1	IC2001-1473		BARE P.C. BOARD	1
2	SA-9335	IC1	555 $\frac{1}{2}$ CONTINUOUSLY VARIABLE SLOPE DELTA MODULATOR	1
3	SA-9321	IC2, IC3	1458 DUAL OP-AMP.	2
4				
5				
6				
7				
8				
9				
10	SA-9846	R11	RESISTOR, FC, 2.0 K OHM 5% 1/4 WATT	1
11	SB-8817	R1, R2, R3, R4, R14, R16	RESISTOR, FC, 10 K OHM 10% 1/4 WATT	6
12	SA-9824	R5	RESISTOR, FC, 15 K OHM 5% 1/4 WATT	1
13	SA-9333	R6	RESISTOR, FC, 15 K OHM 5% 1/4 WATT	1
14	SA-9342	R7	RESISTOR, FC, 3.5 K OHM 5% 1/4 WATT	1
15	SA-9185	R8	POTENTIOMETER, 5 K OHM	1
16	SA-9324	R9, R10, R15, R17	RESISTOR, FC, 27 K OHM 10% 1/4 WATT	6
17	SB-8997	R12, R13	RESISTOR, FC, 2.7 K OHM 10% 1/4 WATT	2
18	SA-8772	R17	RESISTOR, FC, 15 K OHM 5% 1/4 WATT	1
19	SA-9350	C4	CAPACITOR, .047 MFD. 20% 50 VOLT	1
20	SA-9350	C6	CAPACITOR, CERAMIC, 180 PFD. 5% 100 VOLT	1
21	SA-9980	C3, C5, C1, C8, C9, C13 THRU C16	CAPACITOR, CERAMIC, .01 MFD. 50% 20% 50 VOLT	9
22	SA-9031	C1	CAPACITOR, TANTALUM, 1 MFD. 20% 25 VOLT	1
23	SA-9347	C2	CAPACITOR, CERAMIC, 1800 PFD. 5% 50 VOLT	1
24	SA-9343	C12	CAPACITOR, ELECTROLYTIC, 10 MFD. 20% 20 VOLT LOW LEAK	1
25	SA-9343	C11	CAPACITOR, CERAMIC, 4100 PFD. 5% 50 VOLT	1
26	SA-9346	C10	CAPACITOR, CERAMIC, 1200 PFD. 5% 50 VOLT	1
27	SA-9004	J1	24 PIN SOCKET	4
28	SA-9352	J1	RIBBON CABLE ASSEM.	1

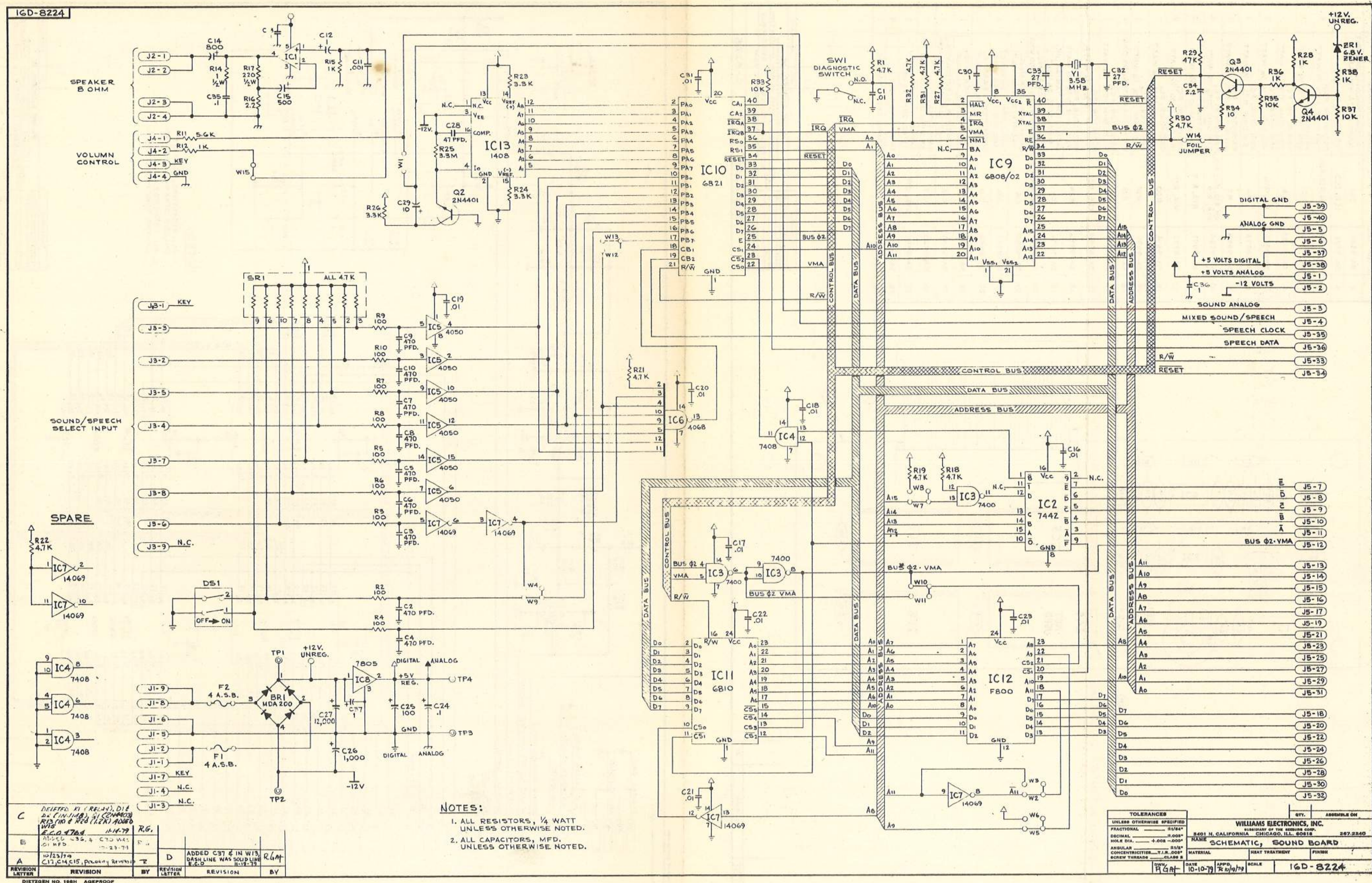


REVISION	REVISION	BY
1	10-6-79	RG

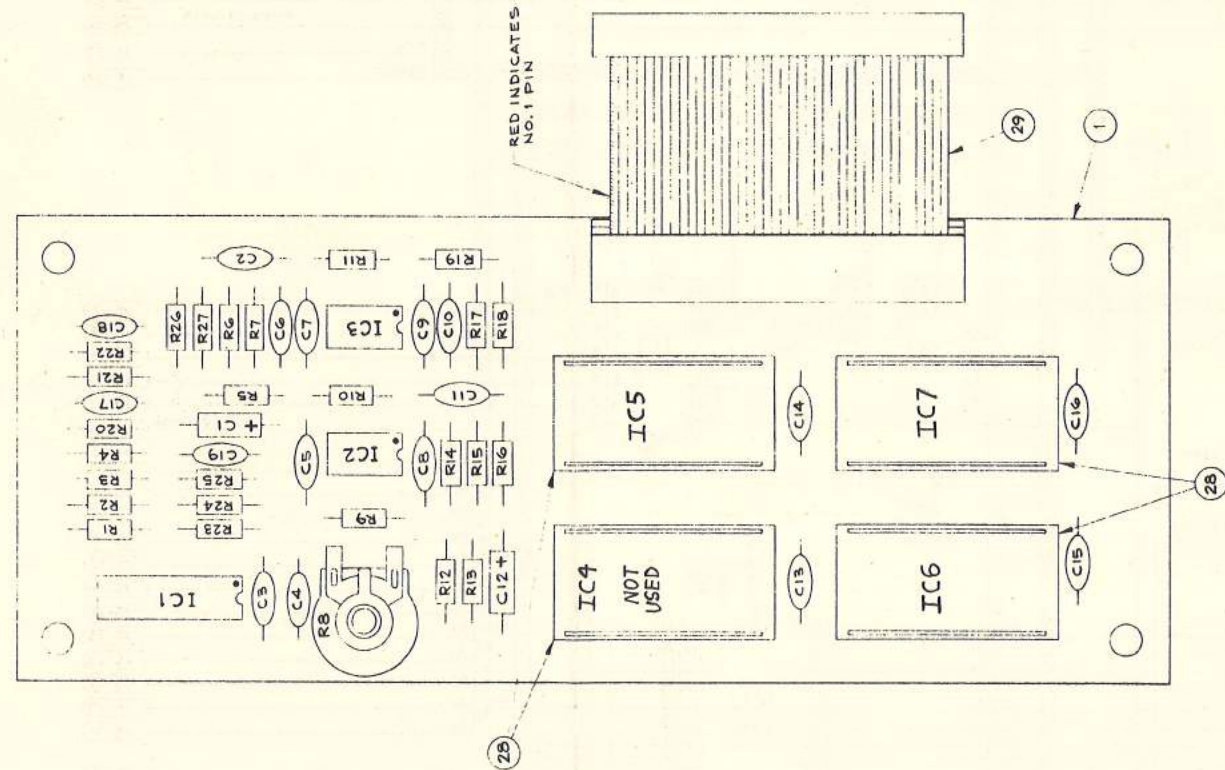
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FRACTIONAL	1/100	WILLIAMS ELECTRONICS, INC.
DECIMAL	1/100	3101 N. CALIFORNIA CHICAGO, ILL. 60618
HOLE DIA.	1/16	NAME SPEECH MODULE ASSEM.
ANGULAR DIMENSIONS	1/16	MATERIAL
SCREEN THICKNESS	1/16	HEAT TREATMENT
FINISH		PRIME
REV.	21	C-8228



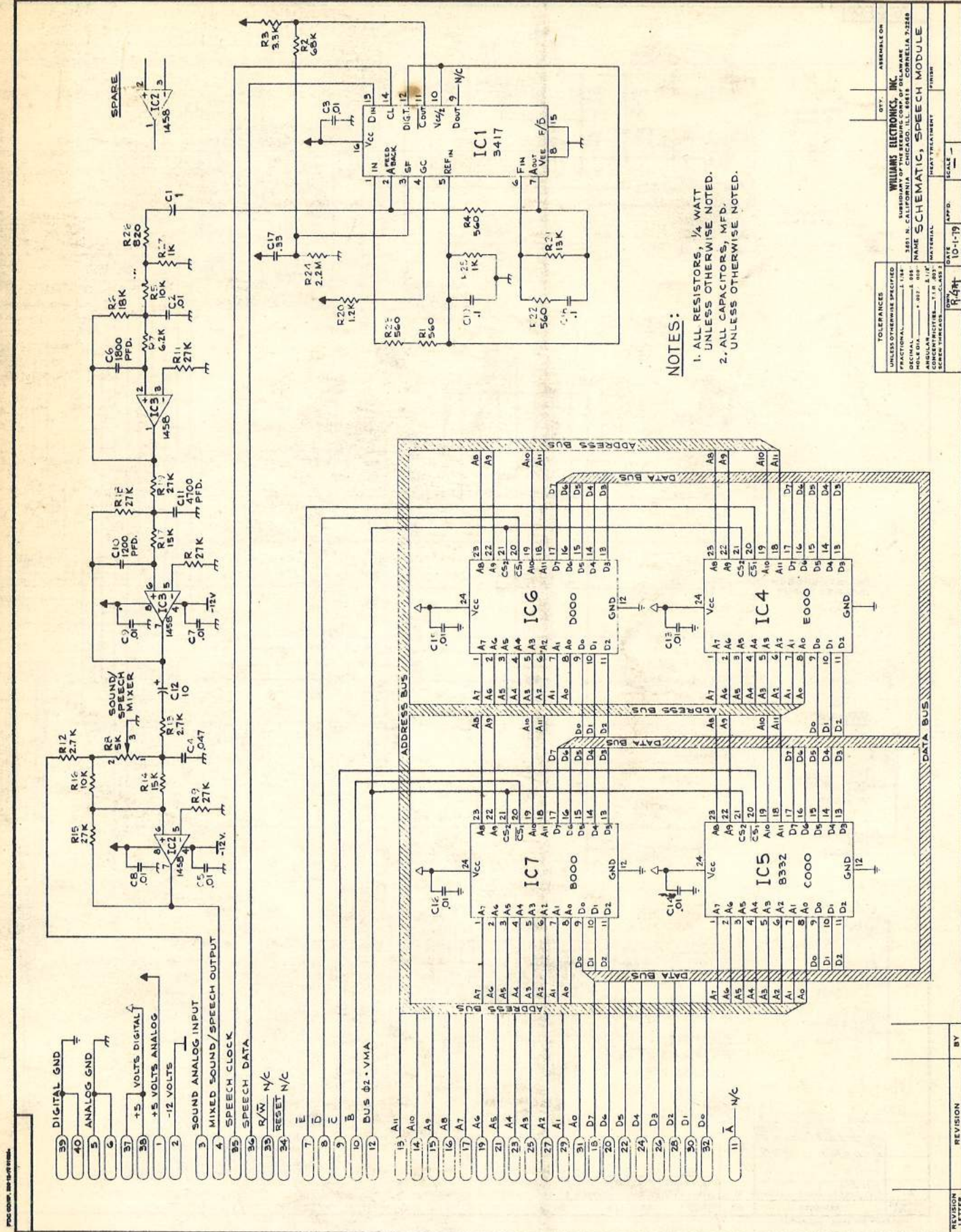
TOLERANCES	UNLESS OTHERWISE SPECIFIED	ASSEMBLY ON
FRACTIONAL	1/100	WILLIAMS ELECTRONICS, INC.
DECIMAL	1/100	3101 N. CALIFORNIA CHICAGO, ILL. 60618
HOLE DIA.	1/16	NAME SCHEMATIC, SPEECH MODULE
ANGULAR DIMENSIONS	1/16	MATERIAL
SCREEN THICKNESS	1/16	HEAT TREATMENT
FINISH		PRIME
REV.	21	C-8228



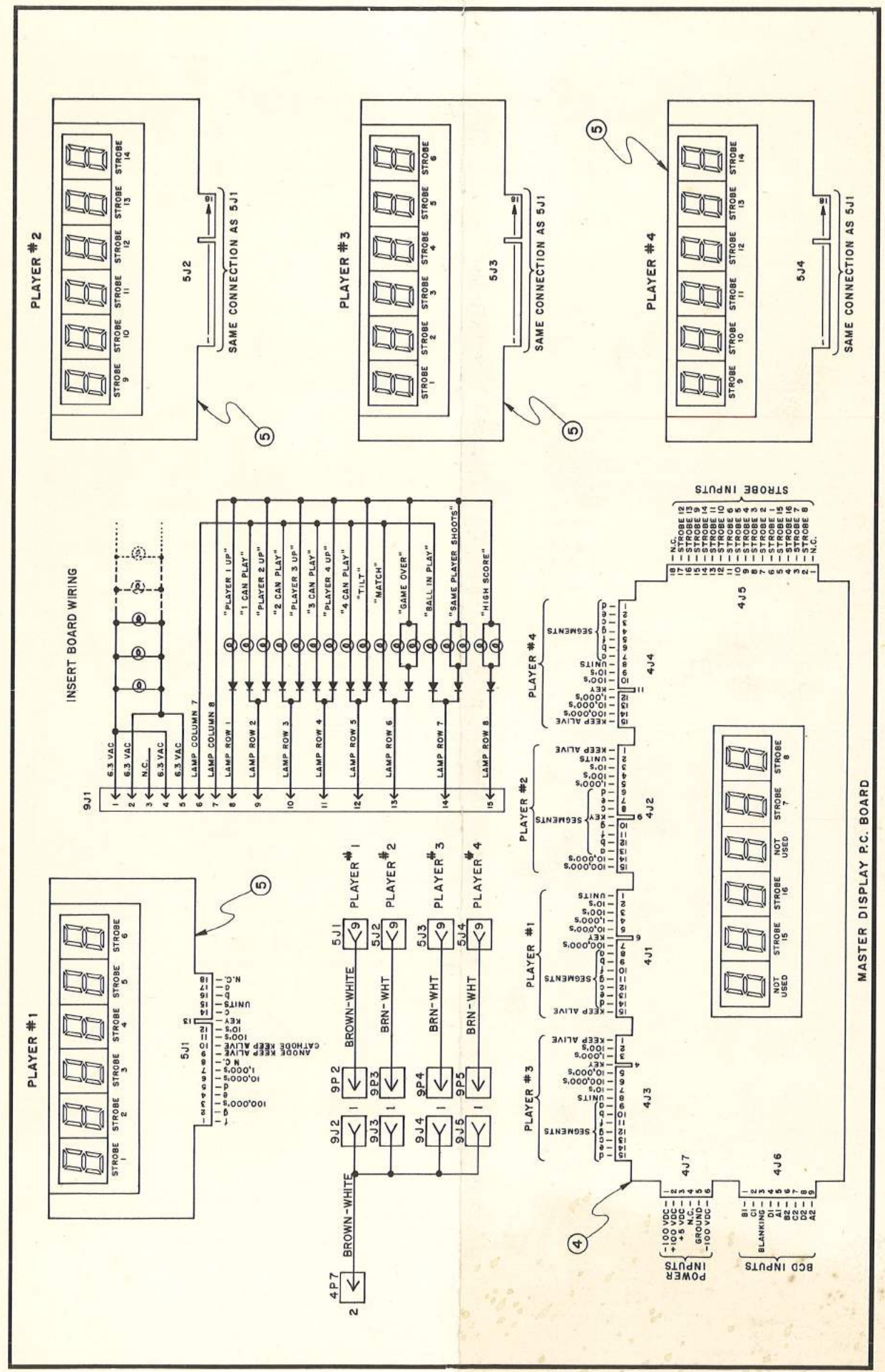
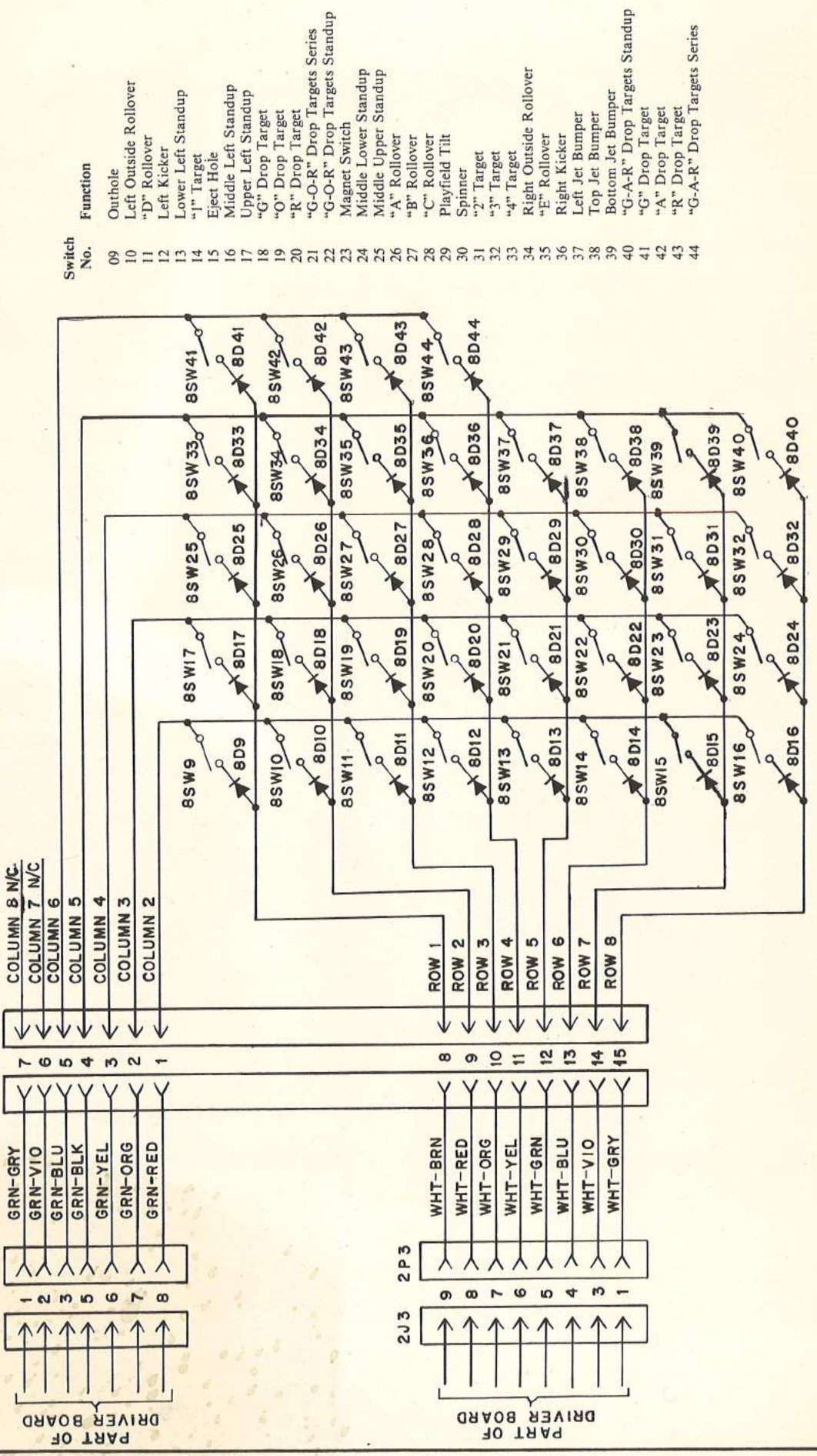
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQD. NO.
1	IC-2001-148-2		BARE P.C. BOARD	1
2	5A-9334	IC1	3417 CONTINUOUSLY VARIABLE SLOPE DELTA MODULATOR	1
3	5A-9321	IC2, IC3	1458 DUAL OP-AMP.	2
4	5A-8992	R1, R4, R22, R23	RESISTOR, FC, 1/4 WATT	4
5	5A-8776	R2	RESISTOR, FC, 1/4 WATT	1
6	5A-8983	R3	RESISTOR, FC, 1/4 WATT	1
7	5A-8817	R5, R16	RESISTOR, FC, 1/4 WATT	2
8	5A-8773	R6	RESISTOR, FC, 1/4 WATT	1
9	5A-9353	R7	RESISTOR, FC, 1/4 WATT	1
10	5A-9324	R9, R10, R11, R15, R18, R19	RESISTOR, FC, 1/4 WATT	6
11	5A-8997	R12, R13	RESISTOR, FC, 1/4 WATT	2
12	5A-8772	R14, R17	RESISTOR, FC, 1/4 WATT	2
13	5A-9314	R20	RESISTOR, FC, 1/4 WATT	1
14	5A-9331	R21	RESISTOR, FC, 1/4 WATT	1
15	5A-9185	R8	POTENTIOMETER, 5K OHM	1
16	5A-9218	R24	RESISTOR, FC, 1/4 WATT	1
17	5A-8984	R25, R27	RESISTOR, FC, 1/4 WATT	1
18	5A-9356	R26	RESISTOR, FC, 1/4 WATT	1
19	5A-9031	C1	CAPACITOR, TANTALUM, 1 MFD, 20% 25 VOLT	1
20	5A-8980	C2, C3, C5, C7, C8, C9, C13 THRU C16	CAPACITOR, CERAMIC, .01 MFD, +80%-20% 50 VOLT	10
21	5A-9030	C4	CAPACITOR, CERAMIC, .047 MFD, 20% 50 VOLT	1
22	5A-9847	C6	CAPACITOR, CERAMIC, 1800 PFD, 5% 50 VOLT	1
23	5A-9346	C10	CAPACITOR, CERAMIC, 1200 PFD, 5% 50 VOLT	1
24	5A-9348	C11	CAPACITOR, CERAMIC, 4100 PFD, 5% 50 VOLT	1
25	5A-9343	C12	CAPACITOR, ELECTROLYTIC, 10 MFD, 20% 25 VOLT LOW LEAK	1
26	5A-9263	C17	CAPACITOR, .33 MFD, 20% 200 VOLT	1
27	5A-8996	C18, C19	CAPACITOR, CERAMIC, .1 MFD, 20% 25 VOLT	2
28	5A-9004		24 PIN SOCKET	4
29	5A-9354	J1	RIBBON CABLE ASSEM	1

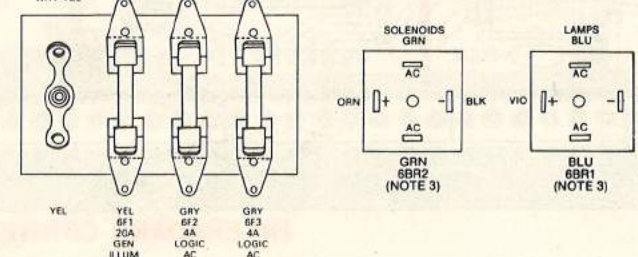
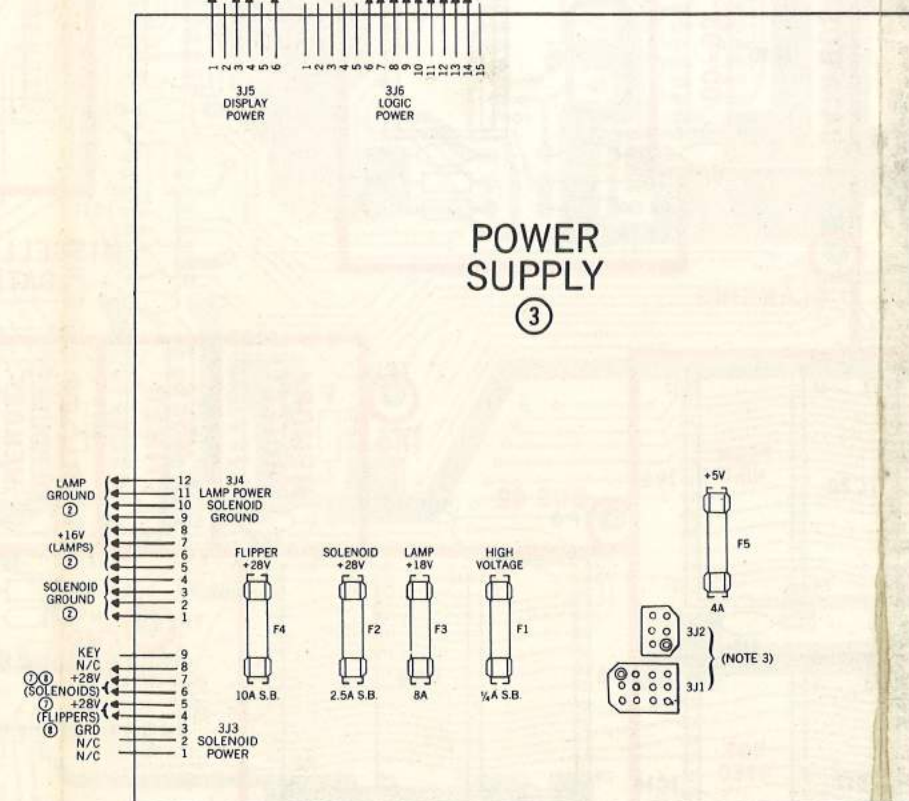
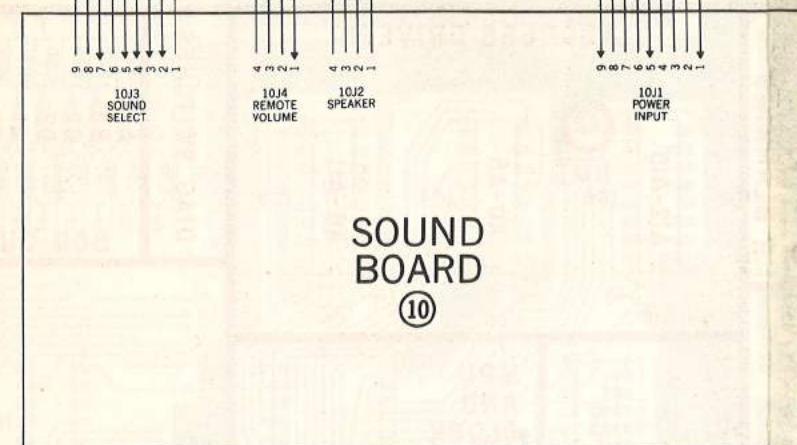
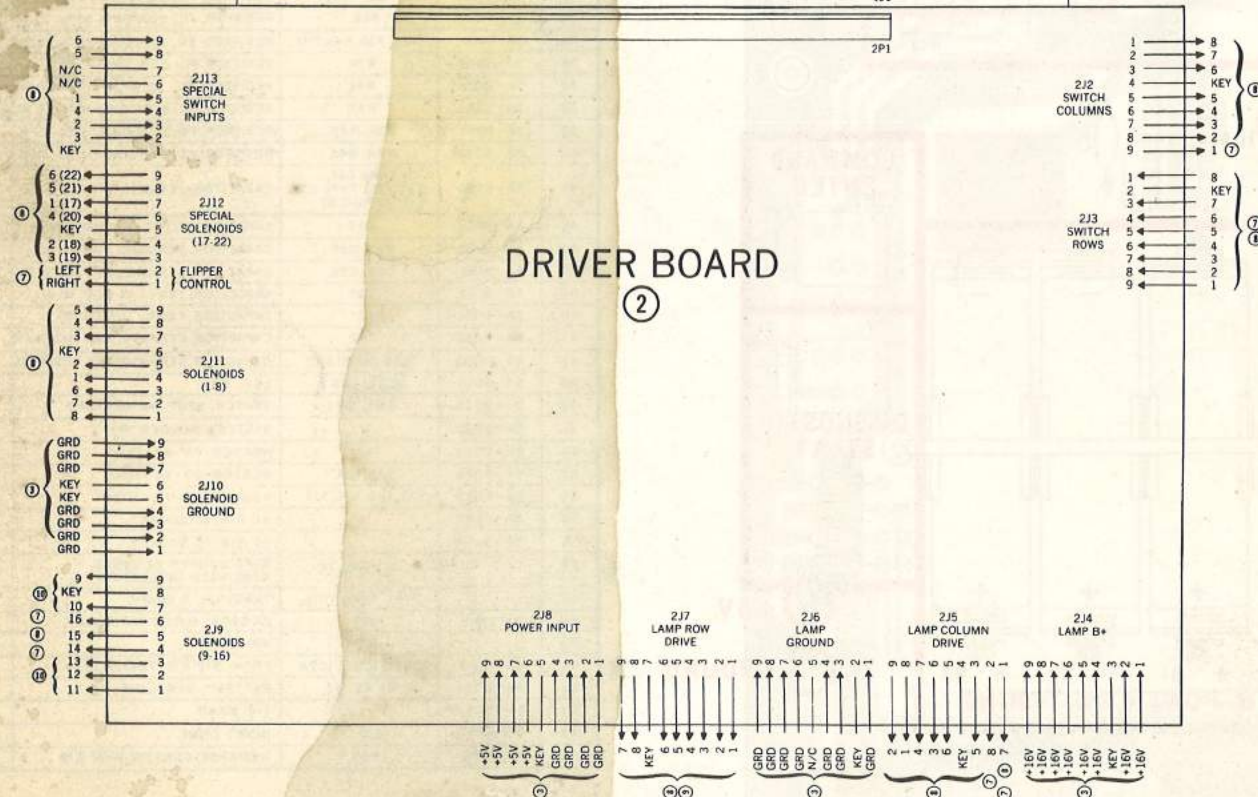


TOLERANCES		UNLESS OTHERWISE SPECIFIED		UNLESS OTHERWISE SPECIFIED		UNLESS OTHERWISE SPECIFIED		UNLESS OTHERWISE SPECIFIED			
RESISTORS		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%			
CAPACITORS		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%			
WELDS		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%		1% - 1% - 1% - 1% - 1%			
ASSEMBLY ON		WILLIAMS ELECTRONICS, INC.		WILLIAMS ELECTRONICS, INC.		WILLIAMS ELECTRONICS, INC.		WILLIAMS ELECTRONICS, INC.			
3801 N. CALIFORNIA, CHICAGO, ILL. 60618		3801 N. CALIFORNIA, CHICAGO, ILL. 60618		3801 N. CALIFORNIA, CHICAGO, ILL. 60618		3801 N. CALIFORNIA, CHICAGO, ILL. 60618		3801 N. CALIFORNIA, CHICAGO, ILL. 60618			
NAME		SPEECH MODULE		NAME		SPEECH MODULE		NAME		SPEECH MODULE	
HEAT TREATMENT		ASSEMBLY		HEAT TREATMENT		ASSEMBLY		HEAT TREATMENT		ASSEMBLY	
DATE		9-28-78		DATE		9-28-78		DATE		9-28-78	
REV.		21		REV.		21		REV.		21	
BY		C-8225		BY		C-8225		BY		C-8225	



GORGAR PLAYFIELD SWITCH WIRING DIAGRAM





NOTES:

1. CONNECTIONS ARE INDICATED BY CIRCLED NUMBERS AS FOLLOWS:
 - ① CPU BOARD
 - ② DRIVER BOARD
 - ③ POWER SUPPLY BOARD
 - ④ MASTER DISPLAY BOARD
 - ⑤ SLAVE DISPLAY BOARD
 - ⑥ BACKBOX
 - ⑦ CABINET
 - ⑧ PLAYFIELD
 - ⑨ INSERT BOARD
 - ⑩ SOUND BOARD
2. PROM #1, #2, AND #3 ARE USED; GAME ROM IS NOT USED.
3. REFER TO POWER WIRING DIAGRAM (PAGE 15) FOR CONNECTIONS TO 3P1, 3P2, 6BR1, 6BR2, 6F1, 6F2, AND 6F3.

Drawing	Page No.
Backbox Wiring Diagram	1
CPU Board Assembly Drawing	2
CPU Board Logic Diagram	3
Driver Board Assembly Drawing	4
Driver Board Logic Diagram (Sheet 1 of 2) ...	5
Driver Board Logic Diagram (Sheet 2 of 2) ...	6
Power Supply Assembly and Schematic Diagrams	7
Sound Board Assembly Drawing	8
Sound Board Logic Diagram	9
D8000 Master Display Board Assembly Drawing (IC Drivers)	10
D8000 Master Display Board Logic Diagram (IC Drivers)	11
D8169 Master Display Board Assembly Drawing (Discrete Drivers)	12
D8169 Master Display Board Logic Diagram (Discrete Drivers)	13
Slave Display Board Assembly and Schematic Diagrams	14
Power and Cabinet Wiring Diagrams	15
Playfield Lamp and Solenoid Wiring Diagrams	16
Playfield Switch and Insert Board Wiring	17

GORGAR

REVISION LETTER	REVISION
D	REVISED AND REDRAWN REVISED DWG. TO CONFORM TO ARTWORK NO. 18-2001-133 LATEST ISSUE NO. 6. R. GAY, 1-8-79
E	ITEM NO. 3, MFG'S PT. NO. WAS IN5991. E.C.O. 4670 R. GAY, 1-31-79

TP1 +12V. UNREG

TP2 NMI

+5V.
0V.
DIAGNOSTIC
SWITCH
DEPRESSED

TP3 MEMORY PROTECT

+5V.
0V.
INTERLOCK
CLOSED INTERLOCK
OPEN

TP4 BLANKING

+5V.
0V.
NORMAL
OPERATION BLANKING

TP5 $\overline{IRQ}$

+5V.
0V.
30μS
1MS

TP6 BUS $\phi 2$

+5V.
0V.
1.1μS

TP7 CMOS RAM B+

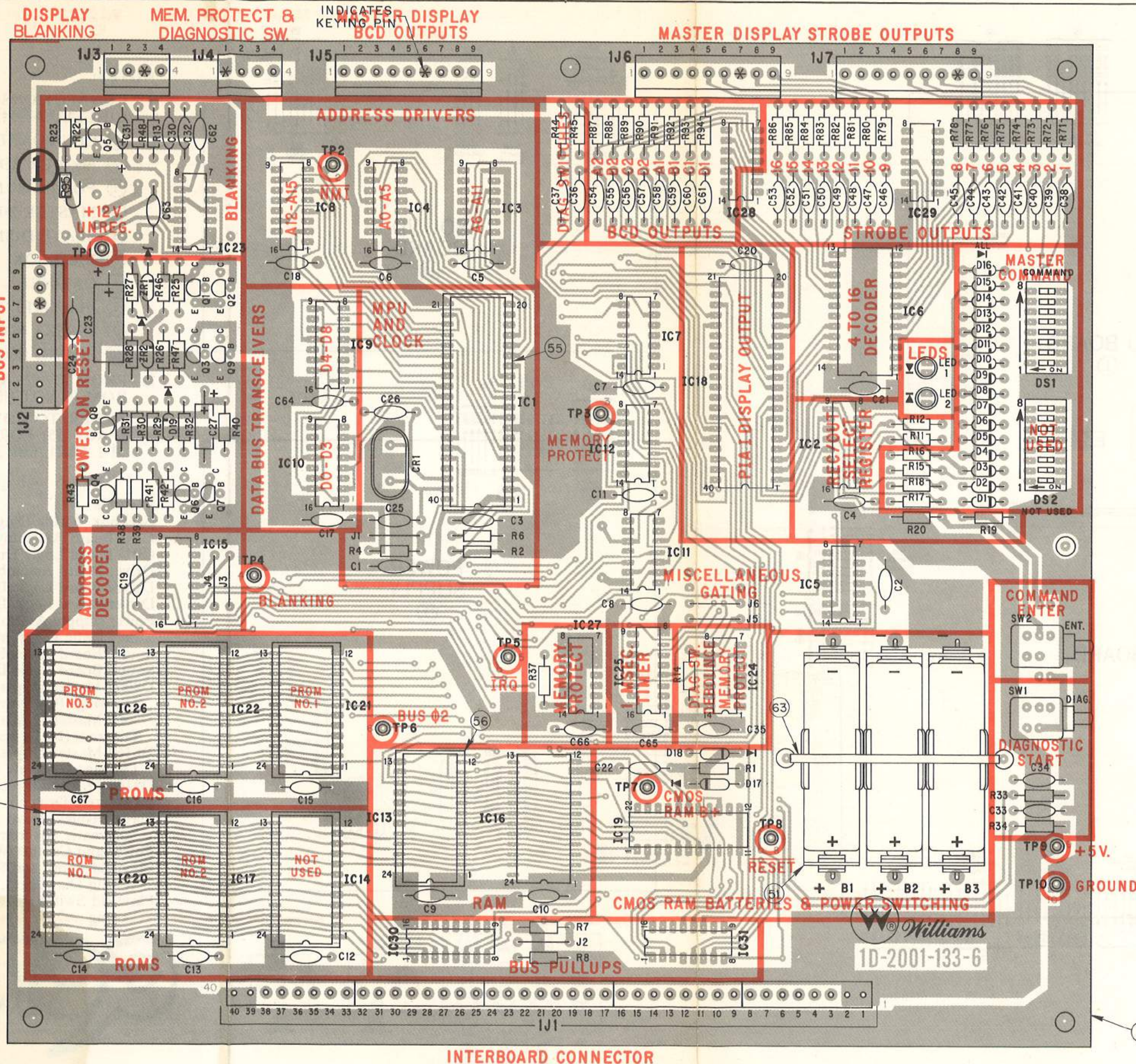
POWER ON 4.3V.
POWER OFF 3.9V.

TP8 RESET

1 SEC.
POWER
TURN ON +5V.
0V.

TP9 +5V.

TP10 GND



BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	18-2001-133-6		BARE P.C. BOARD	1
2	5A-8990	IC2, IC9, IC10	8T28 QUAD BUFFER/RECEIVER	3
3	5A-8989	IC3, IC4, IC8	8T97 HEX. TS BUFFER	3
4	5A-9010	IC6	74154 4 TO 16 DECODER	1
5	5A-9013	IC7	7404 HEX. INVERTER	1
6	5A-9235	IC11	74LS10 TRIPLE 3 INPUT NAND	1
7	5A-8973	IC12	7408 QUAD AND	1
8	5A-9003	IC13, IC16	MC6810 RAM	2
9	5A-9246	IC15	74LS139 DUAL 2 TO 4 LINE DECODER	1
10	5A-9004	IC17	ROM 2K X 8 LOWER	1
11	5A-8972	IC18	MC6820 PIA	1
12	5A-9017	IC19	CMOS RAM 5101	1
13	5A-9000	IC20	ROM 2K X 8 UPPER	1
14	5C-9002	IC23	MC3456/556 DUAL TIMER	1
15	5A-9073	IC24	7400 QUAD 2 INPUT NAND	1
16	5A-9236	IC25	4020 CMOS 14 BIT COUNTER	1
17	5A-9237	IC27	4071 CMOS QUAD 2 INPUT NOR	1
18	5A-9247	IC5	74LS02 QUAD 2 INPUT NOR GATE	1
19	5A-9238	IC28, IC29	13 DIP RESISTOR/PACK, 4.7K OHM	2
20	5A-9239	IC30, IC31	15 DIP RESISTOR/PACK, 4.7K OHM	2
21	5B-9025	DS1, DS2	8 STN. DIP SWITCH	2
22	5A-9018	ZR1	IN5996 ZENER DIODE	1
23	5A-9240	ZR2	IN5990 ZENER DIODE	1
24	5A-8919	D1 THRU D17, D19	1N4148 DIODE, SILICON	19
25	5C-8938	Q1, Q2, Q3, Q6 THRU Q9	2N4401 TRANSISTOR	9
26	5C-9016	Q4, Q5	2N4403 TRANSISTOR	2
27	5A-9020	CR1	CRYSTAL, 3.58 MHZ	1
28	5B-8984	R20, R25, R26, R48, R71 THRU R94	RESISTOR, FC, 1K OHM 10% 1/4 W.	28
29	5B-8983	R2, R6, R7, R8, R23, R30	RESISTOR, FC, 3.3K OHM 10% 1/4 W.	6
30	5B-8991	R4, R13 THRU R19, R33, R34, R41	RESISTOR, FC, 4.7K OHM 10% 1/4 W.	11
31	5A-9033	R1	RESISTOR, FC, 680 OHM 5% 1/4 W.	1
32	5B-9036	R11, R12, R42	RESISTOR, FC, 100 OHM 10% 1/4 W.	3
33	5B-9113	R22, R40	RESISTOR, FC, 33K OHM 5% 1/4 W.	2
34	5B-9034	R27, R28,	RESISTOR, FC, 10K OHM 10% 1/4 W.	2
35	5A-9241	R29, R38, R46, R47	RESISTOR, FC, 22K OHM 10% 1/4 W.	4
36	5A-8998	R31	RESISTOR, FC, 2.2K OHM 10% 1/4 W.	1
37	5A-9039	R32	RESISTOR, FC, 10 OHM 10% 1/4 W.	1
38	5A-9242	R37	RESISTOR, FC, 300K OHM 10% 1/4 W.	1
39	5A-8997	R39, R43	RESISTOR, FC, 2.7K OHM 10% 1/4 W.	2
40	5B-9083	R44, R45	RESISTOR, FC, 470 OHM 10% 1/4 W.	2
41	5A-8980	C1, THRU C21, C30, C33 THRU C37, C63 THRU C67	CAPACITOR, CERAMIC, 01 MFD. 50V.	30
42	5A-8986	C23	CAPACITOR, ELECT, 100 MFD. 10V.	1
43	5A-8996	C22, C24	CAPACITOR, CERAMIC, 1 MFD. 50V.	2
44	5A-9169	C25, C26	CAPACITOR, CERAMIC, 27 PFD. 1K V.	2
45	5A-9243	C27	CAPACITOR, TANT., 10 MFD. 10V.	1
46	5A-9031	C31	CAPACITOR, TANT., 1 MFD. 25V.	1
47	5A-9030	C32	CAPACITOR, CERAMIC, .047 MFD. 50V.	1
48	5A-9065	C38 THRU C62	CAPACITOR, CERAMIC, 470 PFD. 50V.	25
49	5A-9019	LED1, LED2	LED, RED	2
50	5A-9024	SW1, SW2	SWITCH, SPDT MOMENTARY	2
51	5A-9021		BATTERY HOLDER #171	1
52	5A-9026	IJ1	HEADER 09-64-1083	5
53	5A-9028	IJ3, IJ4	HEADER 09-65-1041	2
54	5A-9027	IJ2, IJ5, IJ6, IJ7	HEADER 09-65-1094	4
55	5A-8985		40 PIN IC SOCKET	1
56	5A-9004		24 PIN IC SOCKET	7
57		J1 THRU J6	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	6
58		-TR1 THRU TP10	TERMINAL #1502-1	10
59	5A-9250	IC1	MC6808 MICROPROCESSOR	1
60		IC14	NOT USED	0
61	5A-9015	IC21, IC22, IC26	PROM 512 X 8 7641/6341	2/3
62	5A-9022	B1, B2, B3	BATTERY, ALKALINE, 1.5V.	3
63	3A-7520-1		TIE WRAP	1
64	5A-9266	D18	IN5817 DIODE	1
65	5A-9086	R95	RESISTOR, 68K OHM 10% 1/4 W.	1

★ - WHEN USED, IC14 REPLACES IC21, IC22, AND IC26.

✱ - J2 THRU J4, R1 & R4 NOT USED.

WILLIAMS ELECTRONICS, INC.

SUBSIDIARY OF XCOR CORPORATION

3401 N. CALIFORNIA CHICAGO, ILL. 60618 CORNELIA 7-2240

PART NAME

CPU BOARD ASSEMBLY

DWN.

R. Gay

DATE

1-11-79

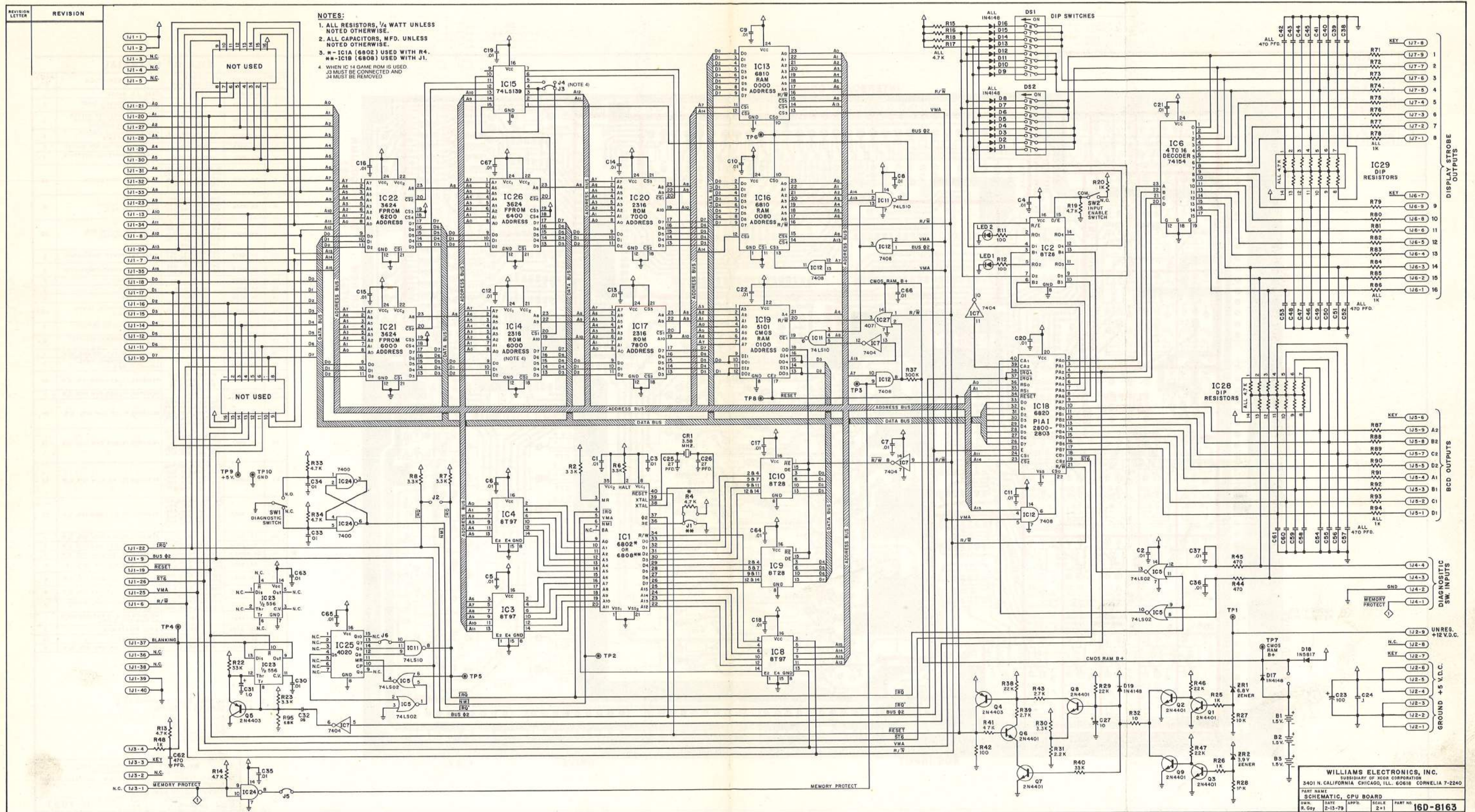
APP'D.

SCALE

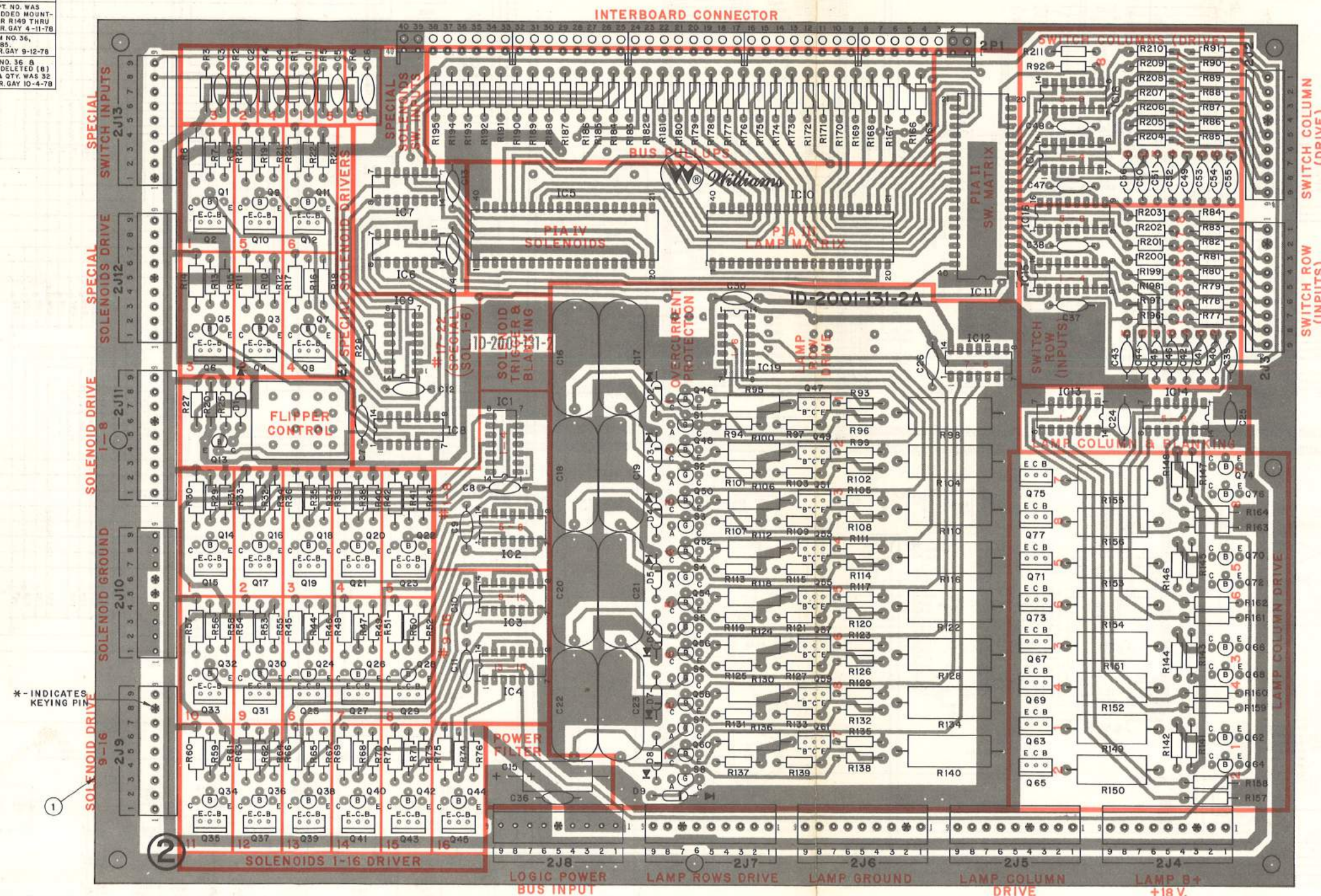
2=1

PART NO.

D-8161



REVISION LETTER	REVISION
C	REVISED AND REDRAWN R.GAY 11-28-77
D	ITEM NO. 28, PT. NO. WAS 5A-8999 & ADDED MOUNT- ING NOTE FOR R149 THRU R156. R.GAY 4-11-78
E	DELETED ITEM NO. 36, PT. NO. 5A-8985 E.C.O. R.GAY 9-12-78
F	ADDED ITEM NO. 36 & ITEM NO. 22, DELETED (8) RESISTORS & QTY. WAS 32 E.C.O. 4624 R.GAY 10-4-78

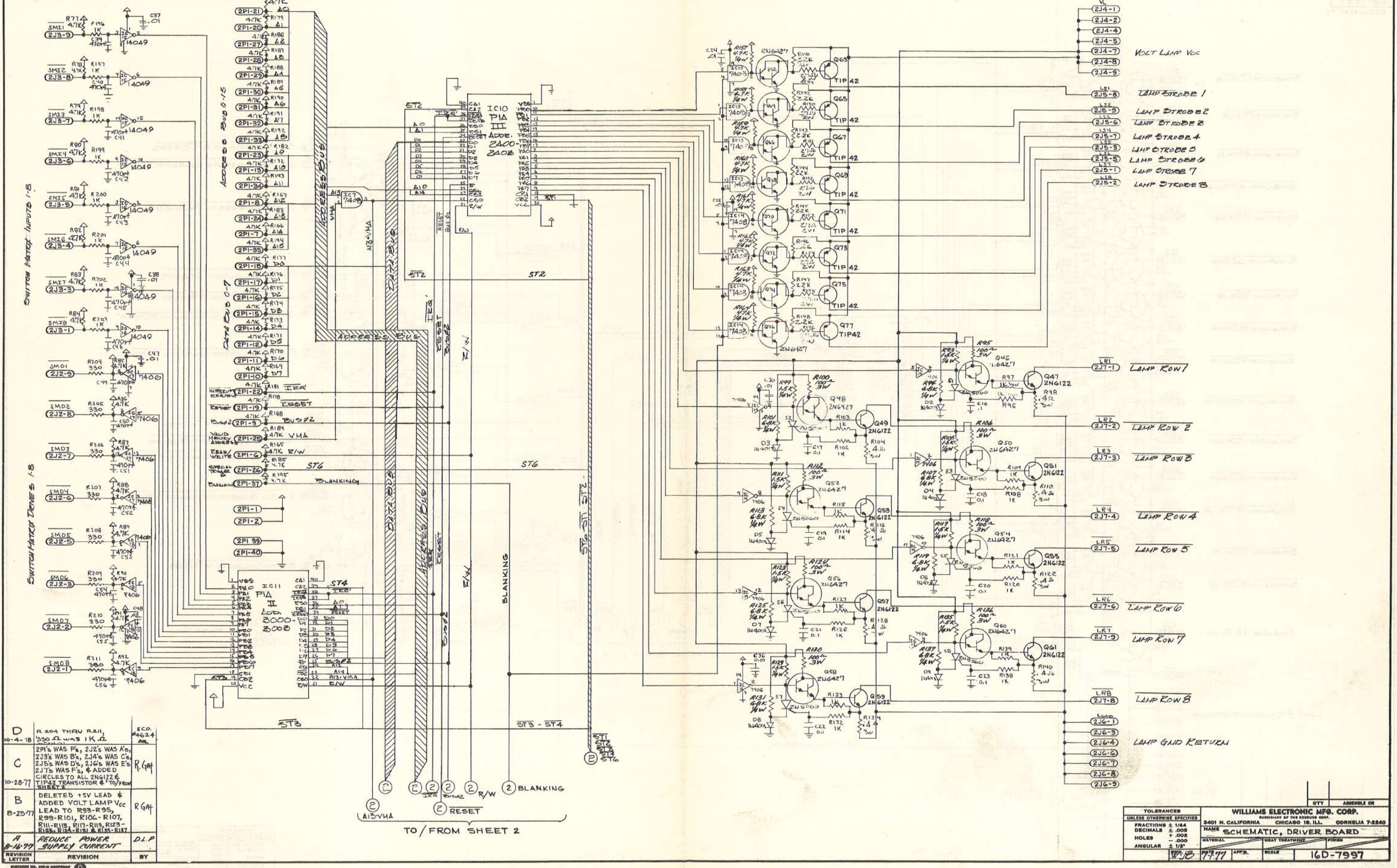


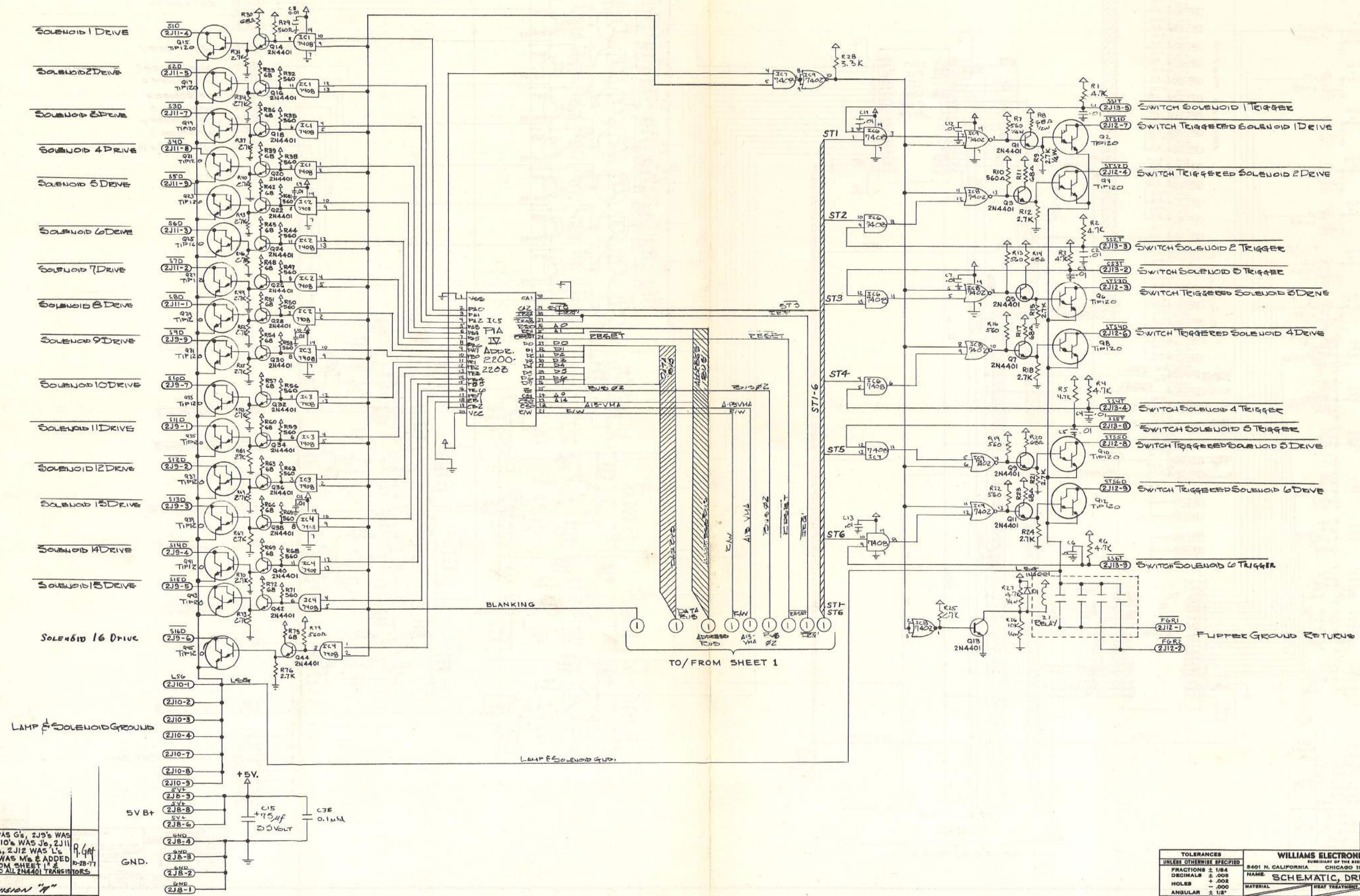
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	1B-2001-131		BARE P.C. BOARD	1
2	5A-8948	IC8, IC9	N7402 QUADRUPLE 2 INPUT POSITIVE NOR GATE	2
3	5A-8974	IC12, IC17, IC18, IC19	N7406 HEX. INVERTER BUFFER DRIVERS W/ OPEN COLLECTOR HIGH VOLTAGE OUTPUTS	4
4	5A-8973	IC1 THRU IC4, IC6, IC7, IC13, IC14	N7408 QUADRUPLE 2 INPUT POSITIVE NOR GATE	8
5	5A-8975	IC15, IC16	MC14049 INVERTING HEX. BUFFER	2
6	5A-8972	IC5, IC10, IC11	MC6820 PERIPHERAL INTERFACE ADAPTER	3
7	5A-8939	Q1, Q3, Q5, Q7, Q9, Q11, Q13, Q14, Q16, Q18, Q20, Q22, Q24, Q26, Q28, Q30, Q32, Q34, Q36, Q38, Q40, Q42, Q44	2N4401 NPN TRANSISTOR	23
8	5A-8976	Q46, Q48, Q50, Q52, Q54, Q56, Q58, Q60, Q62, Q64, Q66, Q68, Q70, Q72, Q74, Q76	2N4247 DARLINGTON NPN TRANSISTOR	16
9	5A-8977	Q2, Q4, Q6, Q8, Q10, Q12, Q14, Q16, Q18, Q21, Q23, Q25, Q27, Q29, Q31, Q33, Q35, Q37, Q39, Q41, Q43, Q45	TIP120 DARLINGTON NPN POWER TRANSISTOR	22
10	5A-8978	Q33, Q35, Q37, Q39, Q41, Q43, Q45, Q71, Q73, Q75, Q77	TIP42 PNP POWER TRANSISTOR	8
11	5A-8979	Q47, Q49, Q51, Q53, Q55, Q57, Q59, Q61	2N6122 NPN POWER TRANSISTOR	8
12	5A-6258	D1	1N4001 DIODE	1
13	5A-8919	D2 THRU D8	1N4148 DIODE	8
14	5A-9014	S1 THRU S8	2N5060 SCR	8
15	5A-8980	C1 THRU C4, C24, THRU C26, C30, C37, C38, C47, C48	CAPACITOR, CERAMIC, 01 MFD. +80 -20% 50 V.	22
16	5A-8995	C16 THRU C23	CAPACITOR, POLYESTER FILM, .1 MFD. 10 V.	7
17	5A-9065	C37 THRU C46, C49 THRU C56	CAPACITOR, CERAMIC, 470 PFD. 20% 50 V.	16
18	5A-8986	C15	CAPACITOR, ELECT., 100 MFD. 10 V.	1
19	5A-8996	C36	CAPACITOR, CERAMIC, 1 MFD. +80 -20% 50 V.	1
20	5A-8991	R1 THRU R6, R27, R77 THRU R92, R107 THRU R195	RESISTOR, FC, 4.7 K OHM 10% 1/4 W	62
21	5A-8983	R27	RESISTOR, FC, 3.3 K OHM 10% 1/4 W	1
22	5A-8984	R96, R97, R102, R103, R108, R109, R114, R115, R121, R122, R126, R127, R132, R133, R138, R139, R196 THRU R203	RESISTOR, FC, 1 K OHM 10% 1/4 W	24
23	5A-8992	R7, R10, R13, R16, R19, R22, R29, R32, R35, R38, R41, R44, R47, R50, R53, R56, R59, R62, R65, R68, R71, R74	RESISTOR, FC, 560 OHM 10% 1/4 W	22
24	5A-8993	R8, R11, R14, R17, R20, R23, R30, R33, R36, R39, R42, R45, R48, R51, R54, R57, R60, R63, R66, R69, R72, R75	RESISTOR, FC, 68 OHM 10% 1/2 W	22
25	5A-8997	R9, R12, R15, R18, R21, R24, R25, R31, R34, R37, R40, R43, R46, R49, R52, R55, R58, R61, R64, R67, R70, R73, R76	RESISTOR, FC, 2.7 K OHM 10% 1/4 W	23
26	5A-8917	R26	RESISTOR, FC, 10 K OHM 10% 1/4 W	1
27	5A-8998	R141 THRU R148	RESISTOR, FC, 2.2 K OHM 10% 1/4 W	8
28	5A-8999-1	R149 THRU R156	RESISTOR, FC, 27 OHM 10% 2 W	8
29	5A-9084	R95, R100, R104, R112, R116, R124, R130, R136	RESISTOR, FC, 100 OHM 10% 3 W	8
30	5A-9085	R93, R99, R105, R111, R117, R123, R129, R135	RESISTOR, FC, 1.5 K OHM 10% 1/4 W	8
31	5A-9086	R94, R101, R107, R113, R119, R125, R131, R137	RESISTOR, FC, 6.8 K OHM 10% 1/4 W	8
32	5A-9037	R98, R104, R110, R116, R122, R128, R134, R140	RESISTOR, WIREWOUND, .4 OHM 10% 3 WATT	8
33	5A-8994	Z1	RELAY - 4 POLE - 5 AMP. CONTACTS 40 OHM COIL 6 V.D.C.	1
34	5A-9066	2P1	8 PIN RECEPTACLE	5
35	5A-9027	2J2 THRU 2J13	9 PIN HEADER	12
36	5A-9001	R204 THRU R211	RESISTOR, FC, 330 OHM 10% 1/4 W	8

★ R149 THRU R156 MUST BE MOUNTED 1/8" ABOVE SURFACE OF BOARD.



WILLIAMS ELECTRONICS, INC. SUBSIDIARY OF ICOR CORPORATION 3401 N. CALIFORNIA CHICAGO, ILL. 60618 CORNELIA 7-2240			
PART NAME DRIVER BOARD ASSEMBLY			
DWN. R. Gay	DATE 8-16-77	APP'D. SCALE 2:1	PART NO. D-7997
DOCUMENT #1			

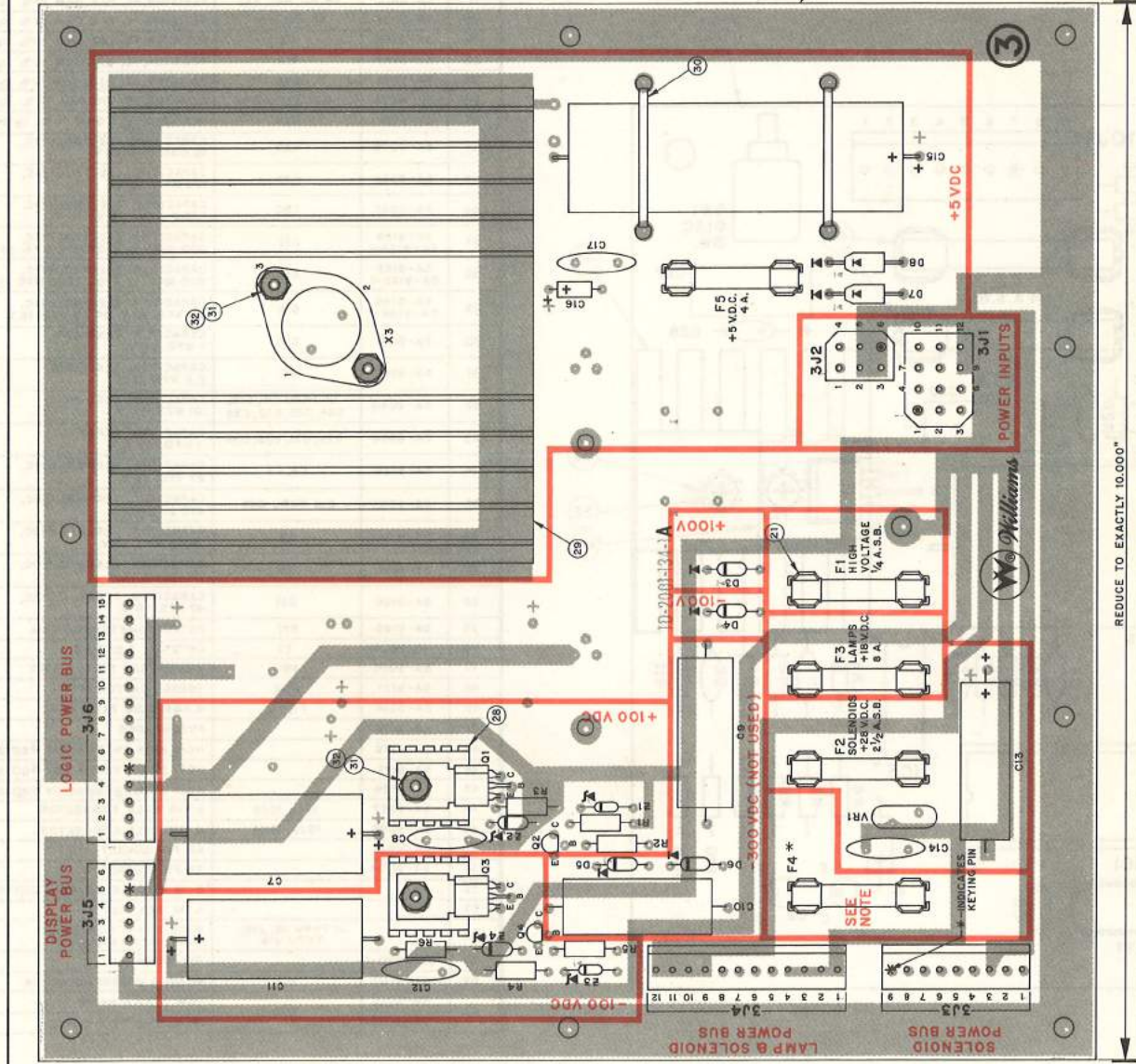




REVISION	REVISION	BY
B	2J8's WAS G's, 2J9's WAS H's, 2J10's WAS J's, 2J11 WAS K's, 2J12 WAS L's 2J13's WAS M's & ADDED 1 TO / FROM SHEET 1 & CIRCLES TO ALL 2N4401 TRANSISTORS	A. G. R.
1	REVISION "A"	

TOLERANCES	UNLESS OTHERWISE SPECIFIED	WILLIAMS ELECTRONIC MFG. CORP.	CHICAGO 18, ILL.	CORNELIA 7-2240
FRACTIONS	± 1/8"	NAME	SCHMATIC, DRIVER BOARD	
DECIMALS	± .005	MATERIAL		
Holes	± .005	HEAT TREATMENT		
ANGULAR	± 1/8°	FINISH		
DATE	1/3/77	SCALE		
16D-7997				

REVISION	REVISION
1	REVISED AND REDRAWN JULY 1978

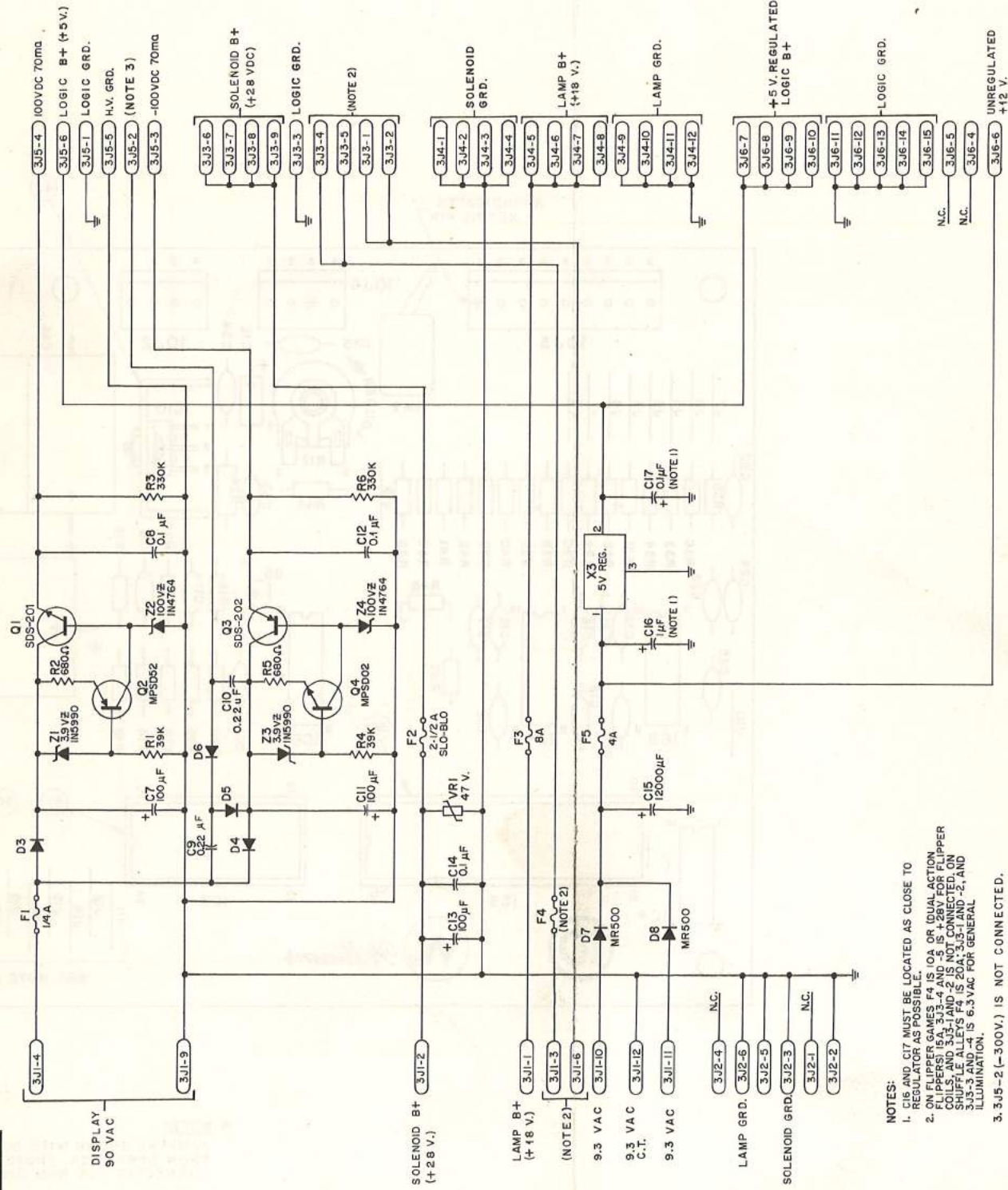


ITEM NO.	PART NO.	PART DESCRIPTION	RECORD NO.
1	10-2001-104	BASE AC BOARD	1
2	5A-9049	5A-9049 POSITIVE VOLTAGE	1
3	5A-9057	5A-9057 NPN TRANSISTOR	1
4	5A-9055	5A-9055 PNP TRANSISTOR	1
5	5A-9058	5A-9058 PNP TRANSISTOR	1
6	5A-9056	5A-9056 PNP TRANSISTOR	1
7	5A-9054	5A-9054 PNP TRANSISTOR	1
8	5A-9045	5A-9045 PNP TRANSISTOR	2
9	5A-9049	5A-9049 POSITIVE VOLTAGE	2
10	5A-9040	5A-9040 POSITIVE VOLTAGE	2
11	5A-9044	5A-9044 POSITIVE VOLTAGE	2
12	5A-9042	5A-9042 POSITIVE VOLTAGE	2
13	5A-9040	5A-9040 POSITIVE VOLTAGE	2
14	5A-9042	5A-9042 POSITIVE VOLTAGE	2
15	5A-9040	5A-9040 POSITIVE VOLTAGE	2
16	5A-9042	5A-9042 POSITIVE VOLTAGE	2
17	5A-9040	5A-9040 POSITIVE VOLTAGE	2
18	5A-9042	5A-9042 POSITIVE VOLTAGE	2
19	5A-9040	5A-9040 POSITIVE VOLTAGE	2
20	5A-9042	5A-9042 POSITIVE VOLTAGE	2
21	5A-9040	5A-9040 POSITIVE VOLTAGE	2
22	5A-9042	5A-9042 POSITIVE VOLTAGE	2
23	5A-9040	5A-9040 POSITIVE VOLTAGE	2
24	5A-9042	5A-9042 POSITIVE VOLTAGE	2
25	5A-9040	5A-9040 POSITIVE VOLTAGE	2
26	5A-9042	5A-9042 POSITIVE VOLTAGE	2
27	5A-9040	5A-9040 POSITIVE VOLTAGE	2
28	5A-9042	5A-9042 POSITIVE VOLTAGE	2
29	5A-9040	5A-9040 POSITIVE VOLTAGE	2
30	5A-9042	5A-9042 POSITIVE VOLTAGE	2
31	5A-9040	5A-9040 POSITIVE VOLTAGE	2
32	5A-9042	5A-9042 POSITIVE VOLTAGE	2
33	5A-9040	5A-9040 POSITIVE VOLTAGE	2
34	5A-9042	5A-9042 POSITIVE VOLTAGE	2
35	5A-9040	5A-9040 POSITIVE VOLTAGE	2
36	5A-9042	5A-9042 POSITIVE VOLTAGE	2
37	5A-9040	5A-9040 POSITIVE VOLTAGE	2

NOTE:
F4 IS FLIPPER FUSE ON FLIPPER
GAMES AND GENERAL ILLUMINATION
ON SHUFFLE ALLEYS.

WILLIAMS ELECTRONICS, INC.	
3401 N. CALIFORNIA, CHICAGO, ILL. 60618 CORNELIA 7-2400	
POWER SUPPLY BOARD ASSEMBLY	
DATE	5-10-79
BY	D-7999
REVISION	1
DOCUMENT #	16C-7999

16C-7999

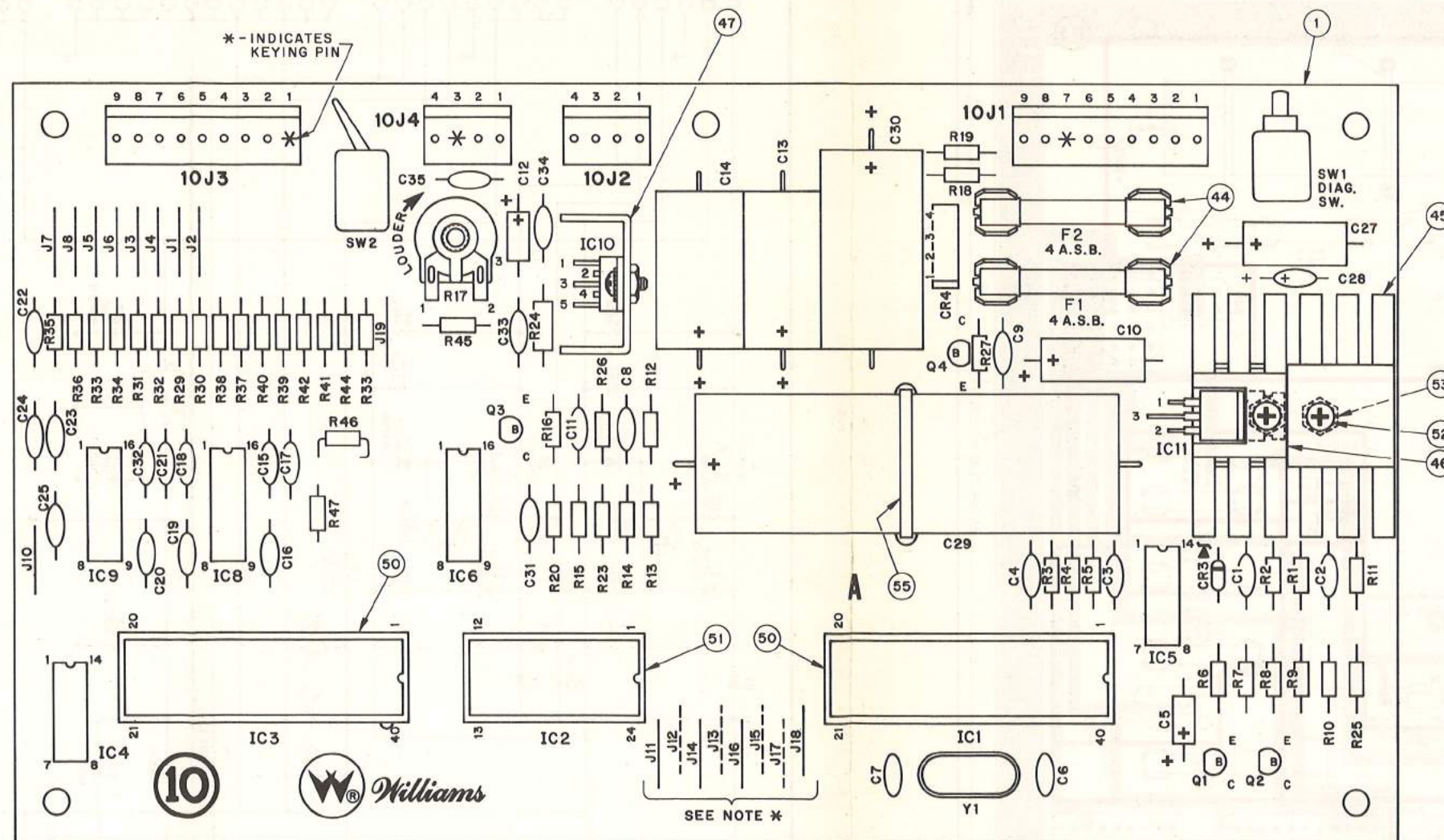


- NOTES:
- C16 AND C17 MUST BE LOCATED AS CLOSE TO REGULATOR AS POSSIBLE.
 - ON FLIPPER GAMES F4 IS 10A OR EQUAL ACTION COILS AND 3J3-1 AND -2 IS NOT CONNECTED ON SHUFFLE ALLEYS F4 IS 20A; 3J3-1 AND -2, AND 3J5-3 AND -4 IS 6.3VAC FOR GENERAL ILLUMINATION.
 - 3J5-2 (-300V.) IS NOT CONNECTED.
 - UNLESS OTHERWISE INDICATED ALL RESISTORS ARE 1/2 W. AND ALL DIODES ARE TYPE 1N4004.

REVISION	REVISION	BY
A	REDRAWN WAS D SIZE. F4 AND NOTES 2,3,4 ADDED	5-9-79

TOLERANCES	UNLESS OTHERWISE SPECIFIED
FRACTIONAL	± 1/16"
DECIMAL	± .005"
HOLE DIA	± .002" - .003"
SCREW THREADS	± .001" - .002"
FINISH	HEAT TREATMENT
DATE	5-10-79
BY	D-7999
REVISION	1
DOCUMENT #	16C-7999

WILLIAMS ELECTRONICS, INC.	ASSEMBLY ON
SUBSIDIARY OF THE SEARS, ROEBUCK & CO. CHICAGO, ILL. 60618 CORNELIA 7-2400	
NAME	SCHEMATIC, POWER SUPPLY BOARD
MATERIAL	
DATE	5-10-79
BY	D-7999
REVISION	1
DOCUMENT #	16C-7999



* NOTE:
JUMPERS SHOWN WITH SOLID LINES CONNECTED FOR
PROM OPERATION. THOSE SHOWN WITH DASHED LINES
CONNECTED FOR ROM OPERATION.

BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	1B-2001-137		BARE P.C. BOARD	1
2	5A-8972	IC3	MC6820 PIA	1
3	5A-9154	IC4	4068 8 INPUT NAND GATE	1
4	5A-9155	IC5	4011 QUAD 2 INPUT NAND GATE	1
5	5A-9152	IC6	MC1408 D/A CONVERTER	1
6	5A-9153	IC8, IC9	4050 BUFFER	2
7	5A-9156	IC10	TD2002V AUDIO AMPLIFIER	1
8	5A-9157	IC11	7805 5 VOLT REG. W/T0220 CASE	1
9	5C-8938	Q1, Q2, Q3	2N4401 NPN TRANSISTOR	3
10	5A-8976	Q4	2N6427 NPN TRANSISTOR	1
11	5A-9018	CR3	1N5996 5.6 V ZENER DIODE	1
12	5A-9158	CR4	MDA200 / 3N253 BRIDGE RECTIFIER	1
13	5B-8991	R1, R2, R14, R23, R37 THRU R45	RESISTOR, FC, 4.7 K OHM 10% 1/4 W	13
14	5B-8983	R3, R4, R5, R15, R16, R20	RESISTOR, FC, 3.3 K OHM 10% 1/4 W	6
15	5A-8984	R9, R10, R25, R27	RESISTOR, FC, 1 K OHM 10% 1/4 W	5
16	5B-9039	R7	RESISTOR, FC, 10 OHM 10% 1/4 W	1
17	5B-8817	R8, R11, R46, R47	RESISTOR, FC, 10 K OHM 10% 1/4 W	4
18	5B-9035	R6, R12, R13	RESISTOR, FC, 47 K OHM 10% 1/4 W	3
19	5A-9160	R18	RESISTOR, FC, 220 OHM 10% 1/4 W	1
20	5A-9161	R19	RESISTOR, FC, 2.2 OHM 10% 1/4 W	1
21	5A-9179	R26	RESISTOR, FC, 3.3 M OHM 10% 1/4 W	1
22	5B-9036	R29 THRU R36	RESISTOR, FC, 100 OHM 10% 1/4 W	8
23	5A-9181	R24	RESISTOR, FC, 1 OHM 10% 1/2 W	1
24	5A-9046	C29	CAPACITOR, ELECTROLYTIC, 12,000 MFD. 16 V.	1
25	5A-8986	C27	CAPACITOR, ELECTROLYTIC, 100 MFD. 10 V.	1
26	5A-8893	C30	CAPACITOR, ELECTROLYTIC, 1,000 MFD. 25 V.	1
27	5A-9164 5A-9164-1	C13	CAPACITOR, ELECTROLYTIC, 500 MFD. 15 V. OR 470 MFD. 25 V.	1
28	5A-9165 5A-9165-1	C14	CAPACITOR, ELECTROLYTIC, 800 MFD. 16 V. OR 1,000 MFD. 16 V.	1
29	5A-9166 5A-9166-1	C10	CAPACITOR, ELECTROLYTIC, 25 MFD. 20 V. OR 33 MFD. 16 V.	1
30	5A-9031	C12	CAPACITOR, TANTALUM, 1 MFD. 25 V.	1
31	5A-9163	C5	CAPACITOR, TANTALUM, 2.2 MFD. 15 V.	1
32	5A-8980	C1 THRU C4, C15, C24, C25, C32, C35	CAPACITOR, CERAMIC, .01 MFD. 50 V.	9
33	5A-8996	C28, C31, C33, C34	CAPACITOR, CERAMIC, .1 MFD. 50 V.	4
34	5A-9169	C6, C7	CAPACITOR, CERAMIC DISC, 27 PFD. 1K V.	2
35	5A-9065	C16 THRU C23	CAPACITOR, CERAMIC DISC, 470 PFD. 50 V.	8
36	5A-9166	C9	CAPACITOR, CERAMIC DISC, .0015 MFD. 1K V.	1
37	5A-9167	C8	CAPACITOR, CERAMIC DISC, .003 MFD. 1K V.	1
38	5A-9180	C11	CAPACITOR, CERAMIC DISC, 47 PFD. 1K V.	1
39	5A-9185	R17	POTENTIOMETER, 5K OHM	1
40	5A-9020	Y1	CRYSTAL, 3.58 MHE.	1
41	5A-9024	SW1	MOMENTARY SWITCH SPDT	1
42	5A-9171	SW2	TOGGLE SWITCH SPDT	1
43	5A-6314	F1, F2	4 AMP. SLOW BLOW FUSE	2
44	5A-9178		FUSEHOLDER	4
45	5A-9172		HEAT SINK THERMALLOY #6072B	1
46	5A-9173		HEAT SINK THERMALLOY #6071B	1
47	5A-9199		HEAT SINK THERMALLOY #6030	1
48	5A-9027	10J1, 10J3	9 PIN MALE CONNECTOR	2
49	5A-9028	10J2, 10J4	4 PIN MALE CONNECTOR	2
50	5A-8985		40 PIN SOCKET	2
51	5A-9004		24 PIN SOCKET	1
52			6-32 X 3/8 BINDER HEAD SCREW	2
53			6-32 HEX. NUT	2
54		J1 THRU J8, J10, THRU J19	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	18
55	5A-7520-1		TIE WRAP	1
56	5A-9150	IC1	MC6802 MICROPROCESSOR	1
57	5A-9015	IC2	7641 512 X 8 PROM	1
	5A-9198		MCM68312 2 K X 8 ROM	1

WILLIAMS ELECTRONICS, INC.			
SUBSIDIARY OF XCOR CORPORATION			
3401 N. CALIFORNIA CHICAGO, ILL. 60618 CORNELIA 7-2240			
PART NAME			
SOUND BOARD ASSEMBLY			
DWN.	DATE	APP'D.	SCALE
R. Gay	12-20-78		2=1
PART NO.			D-8121/25

DIGIT CROSS-REFERENCE

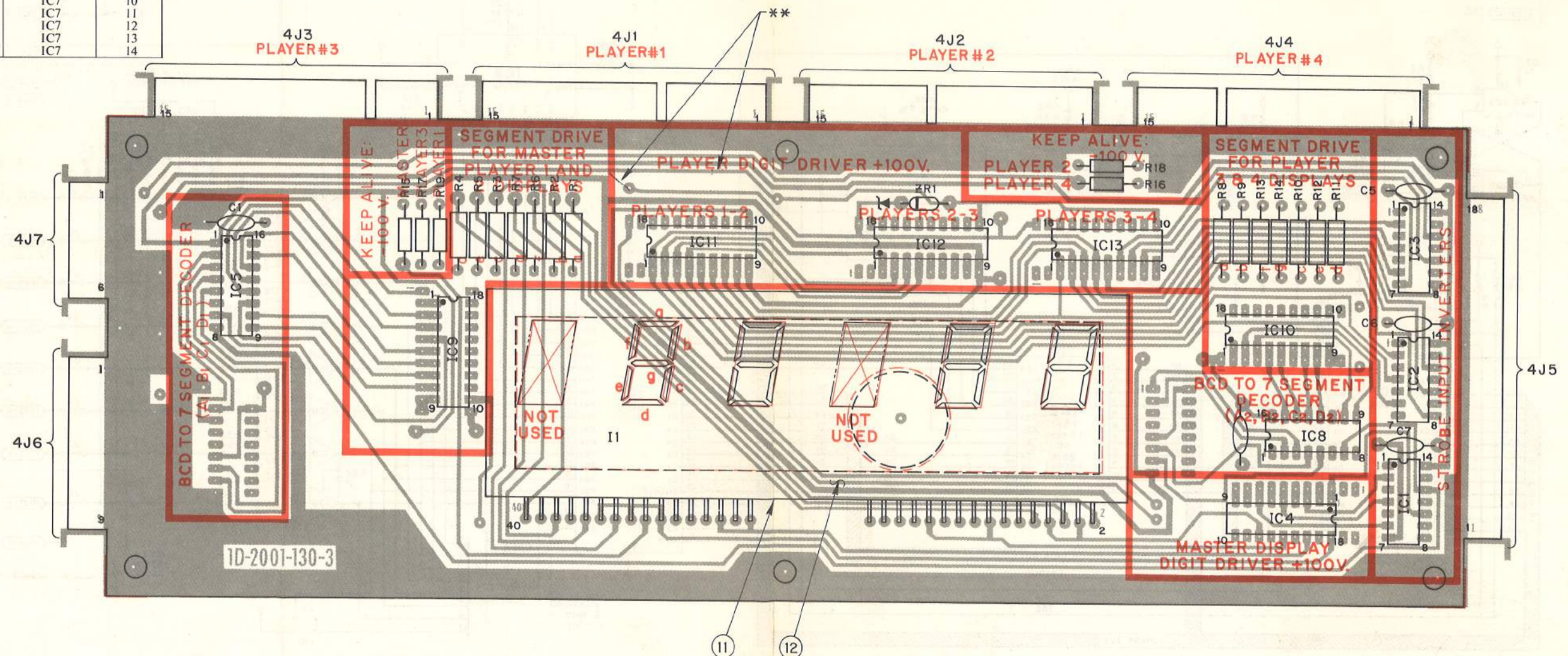
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Master 1 (Left)	IC5	15
Master 2	IC5	16
Master 3	IC5	7
Master 4 (Right)	IC5	8
#1 100,000	IC5	1
#1 10,000	IC5	2
#1 1,000	IC5	3
#1 100	IC5	4
#1 10	IC5	5
#1 Units	IC5	6
#2 100,000	IC5	9
#2 10,000	IC5	10
#2 1,000	IC5	11
#2 100	IC5	12
#2 10	IC5	13
#2 Units	IC5	14
#3 100,000	IC7	1
#3 10,000	IC7	2
#3 1,000	IC7	3
#3 100	IC7	4
#3 10	IC7	5
#3 Units	IC7	6
#4 100,000	IC7	9
#4 10,000	IC7	10
#4 1,000	IC7	11
#4 100	IC7	12
#4 10	IC7	13
#4 Units	IC7	14

NOTES:

- * - IF IC11, IC12, IC13 & IC14 ARE DIONICS-512 DRIVERS THEN R1-R14 ARE 5A-9149 RES., F.C., 15K OHM $\pm 10\%$ $\frac{1}{2}$ W. ZR1 NOT USED.
 ** - CUT & JUMP ON +100V. NOT IMPLEMENTED.

BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	ID-2001-130-3		BARE P.C. BOARD	1
2	5A-8971	IC1, IC2, IC3	MC14069 HEX. INVERTER	3
3	5A-8970	IC5, IC8	MC14543 BCD TO SEVEN SEGMENT LATCH/DECODER/DRIVER	2
4	5A-8969	IC9, IC10	UDN-7180 GAS DISCHARGE DISPLAY SEGMENT DRIVER	2
* 5	5A-8968	IC4, IC11, IC12, IC13	UDN-6184 GAS DISCHARGE DISPLAY SEGMENT DRIVER	2
* 6	5B-8981	R1 THRU R14	RESISTOR, FC, 10K OHM 10% $\frac{1}{2}$ W	14
7	5B-8982	R15 THRU R19	RESISTOR, FC, 3 MEG OHM 10% $\frac{1}{4}$ W	5
8	5A-9135	ZR1	1N4740A ZENER DIODE, 10V. 5% 1W	1
9	5A-8980	C1, C4 THRU C7	CAPACITOR, CERAMIC, .01 MFD. 50V.	5
** 10			JUMPER, #22 GA. SOLID WIRE	1
11	5B-8966	I1	6 DIGIT DISPLAY	1
12	23A 6542		DISPLAY MTG. ADHESIVE FORM	1



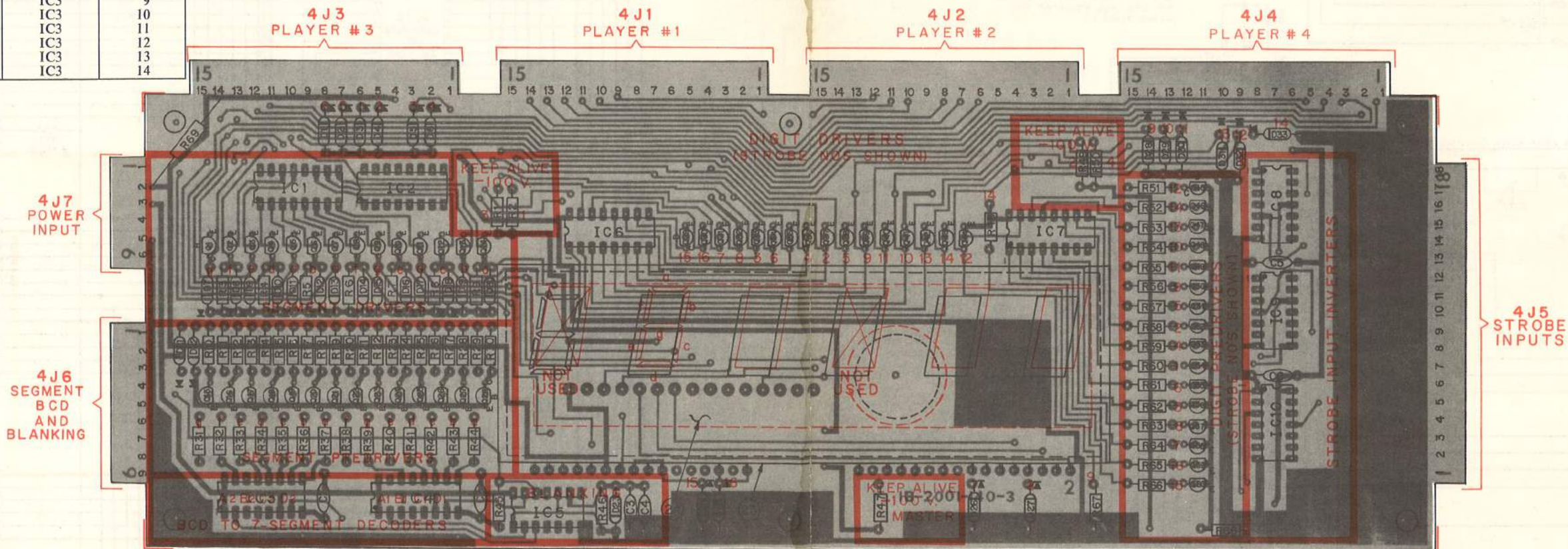
WILLIAMS ELECTRONICS, INC. SUBSIDIARY OF XCOR CORPORATION 3401 N. CALIFORNIA CHICAGO, ILL 60618 CORNELIA 7-2240				
PART NAME	DATE	APP'D.	SCALE	PART NO.
MASTER DISPLAY ASSEMBLY	1-30-79		2=1	D-8000
DWN. R. Gay				

D-8168

DIGIT CROSS-REFERENCE

DIGIT	7-SEGMENT DECODER	STROBE
Master 1 (Left)	IC4	15
Master 2	IC4	16
Master 3	IC4	7
Master 4 (Right)	IC4	8
#1 100,000	IC4	1
#1 10,000	IC4	2
#1 1,000	IC4	3
#1 100	IC4	4
#1 10	IC4	5
#1 Units	IC4	6
#2 100,000	IC4	9
#2 10,000	IC4	10
#2 1,000	IC4	11
#2 100	IC4	12
#2 10	IC4	13
#2 Units	IC4	14
#3 100,000	IC3	1
#3 10,000	IC3	2
#3 1,000	IC3	3
#3 100	IC3	4
#3 10	IC3	5
#3 Units	IC3	6
#4 100,000	IC3	9
#4 10,000	IC3	10
#4 1,000	IC3	11
#4 100	IC3	12
#4 10	IC3	13
#4 Units	IC3	14

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.	ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
17	5A-8774	R45, R67	RESISTOR, FC, 22 K OHM 10% 1/4 W	2	1	1B-2000-140-3		BARE P.C. BOARD	1
18	5A-9035	R10, R12, R13, R15, R16, R18, R19, R21, R22, R24, R25, R27, R26, R30	RESISTOR, FC, 47 K OHM 10% 1/4 W	14	2	5A-9221	IC1	15 DIP RESISTOR/PACK 2.2 K OHM	1
19	5A-9162	R3 THRU R9, R11, R14, R17, R20, R23, R26, R29, R46	RESISTOR, FC, 100 K OHM 10% 1/4 W	15	3	5A-9222	IC2	15 DIP RESISTOR/PACK 1 K OHM	1
20	5A-9218	R1, R2, R47, R49, R50	RESISTOR, FC, 2.2 M OHM 10% 1/4 W	5	4	5A-8970	IC3, IC4	MC14543 7 SEGMENT DRIVER	2
21	5A-8980	C1, C2, C5, C6	CAPACITOR, CERAMIC, .01 MFD. 50V	4	5	5A-9213	IC5	4012 CMOS DUAL 4 INPUT NAND GATE	1
22	5A-9065	C3, C4	CAPACITOR, CERAMIC, 470 PFD. 50V	2	6	5A-9220	IC6	15 DIP RESISTOR/PACK 2.2 K OHM	1
23	58-8966	11	6-DIGIT DISPLAY	1	7	5A-9223	IC7	15 DIP RESISTOR/PACK 10 K OHM	1
24	23A-6542	F1	DISPLAY MOUNTING ADHESIVE FOAM	1	8	5A-9267	IC8, IC9, IC10	4069 LOW PWR. HEX. INVERTER	3
25	5A-9285	R69	RESISTOR, FC, 180 OHM 5% 1/2 W	1	9	5A-9216	Q1 THRU Q14 Q45 THRU Q60	HIGH VOLTAGE NPN TRANSISTOR MPS-A42	30
					10	5A-9217	Q15 THRU Q44	HIGH VOLTAGE PNP TRANSISTOR MPS-A92	30
					11	5A-8785	D1 THRU D20 D24 THRU D33	1N4003 DIODE, SILICON	30
					12	5A-8919	D21, D22, D23	1N4148 DIODE, SILICON	3
					13	5A-9224	R68	RESISTOR, FC, 270 OHM 10% 1/4 W	1
					14	5A-9219	R51 THRU R66	RESISTOR, FC, 8.2 K OHM 10% 1/4 W	16
					15	5A-8817	R48	RESISTOR, FC, 10 K OHM 10% 1/4 W	1
					16	5A-9032	R31 THRU R44	RESISTOR, FC, 12 K OHM 10% 1/4 W	14



WILLIAMS ELECTRONICS, INC.

SUBSIDIARY OF XCOR CORPORATION

3401 N. CALIFORNIA CHICAGO, ILL. 60618 CORNELIA 7-2240

PART NAME

DISCRETE MASTER DISPLAY BD. ASSEM.

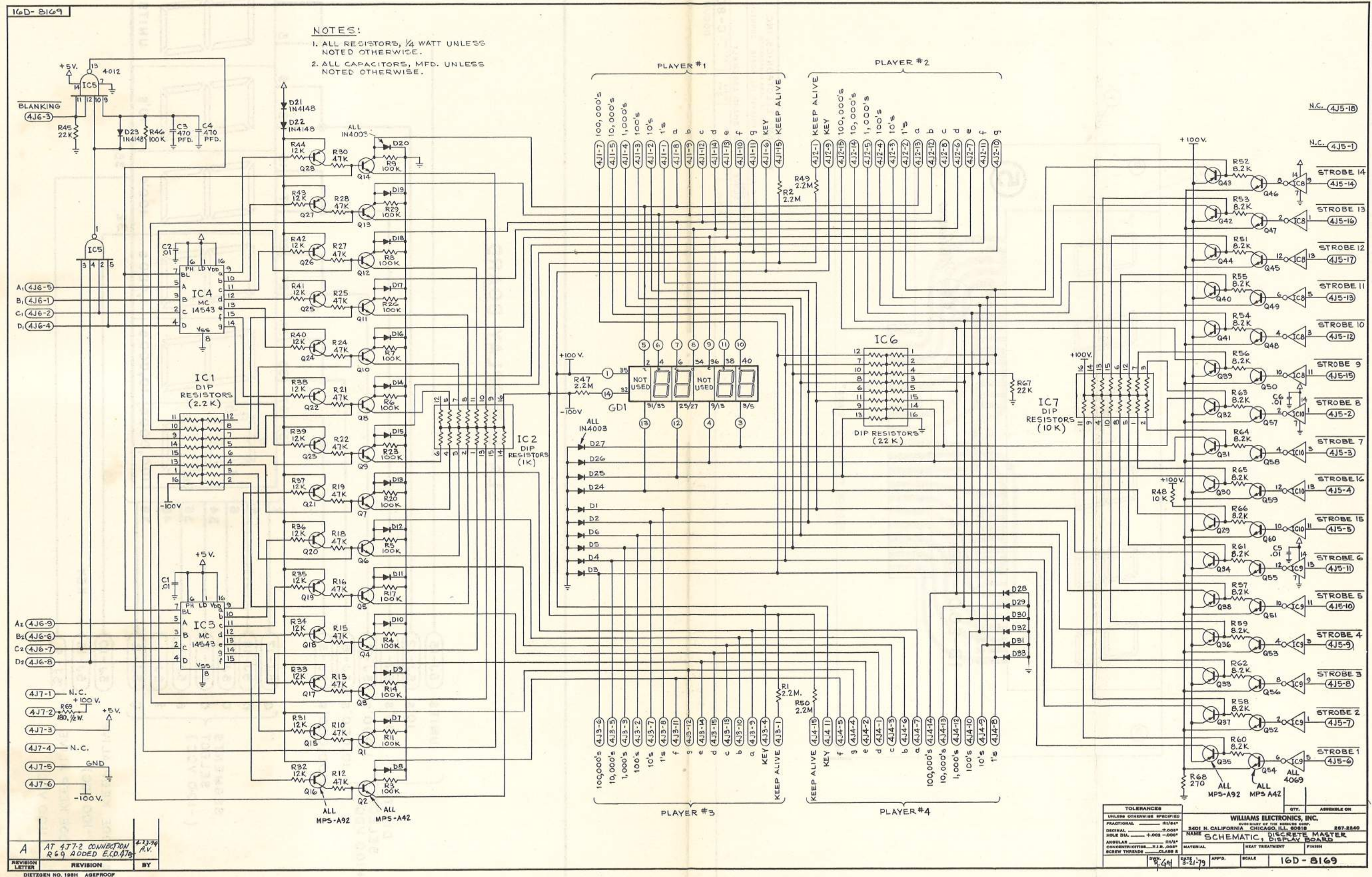
DWN. A.V.

DATE 5-8-79

APP'D.

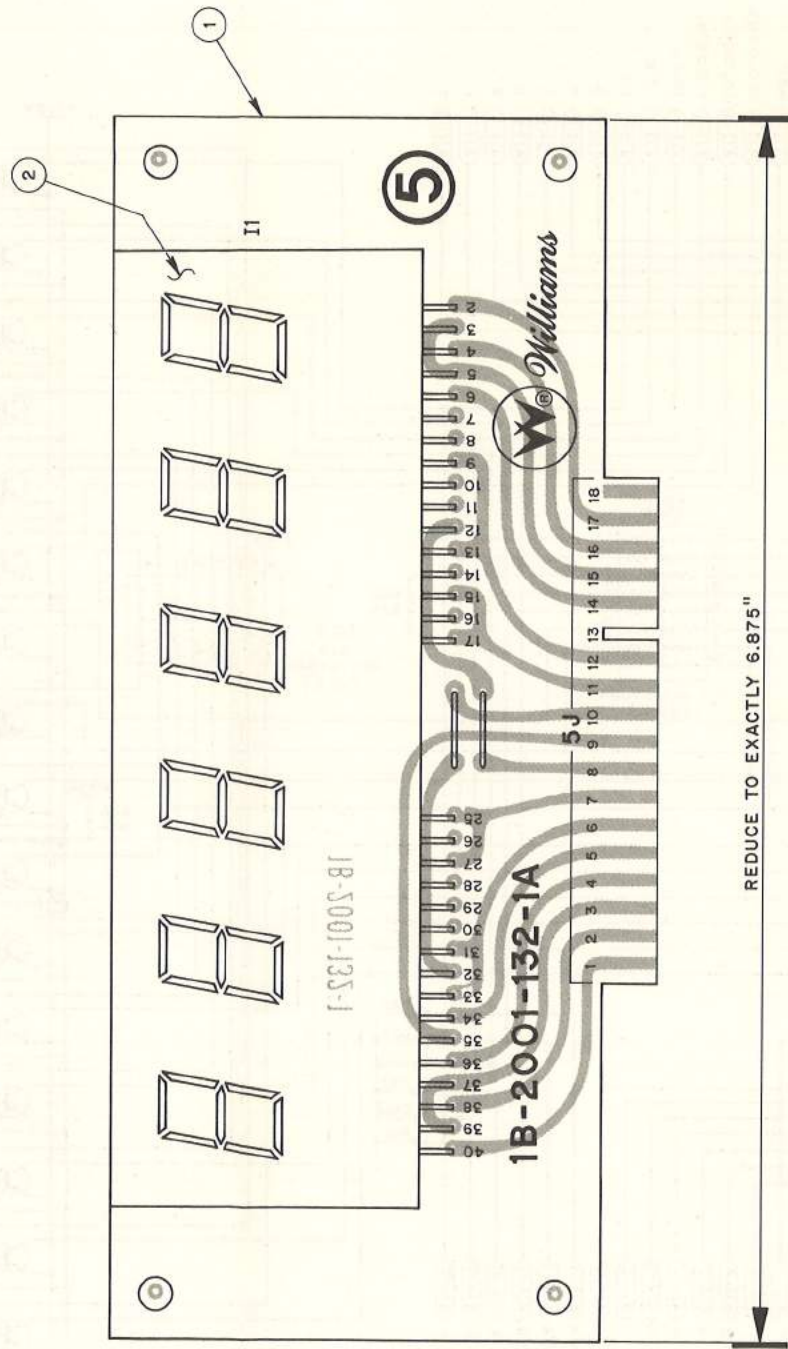
SCALE 2=1

PART NO. D-8168



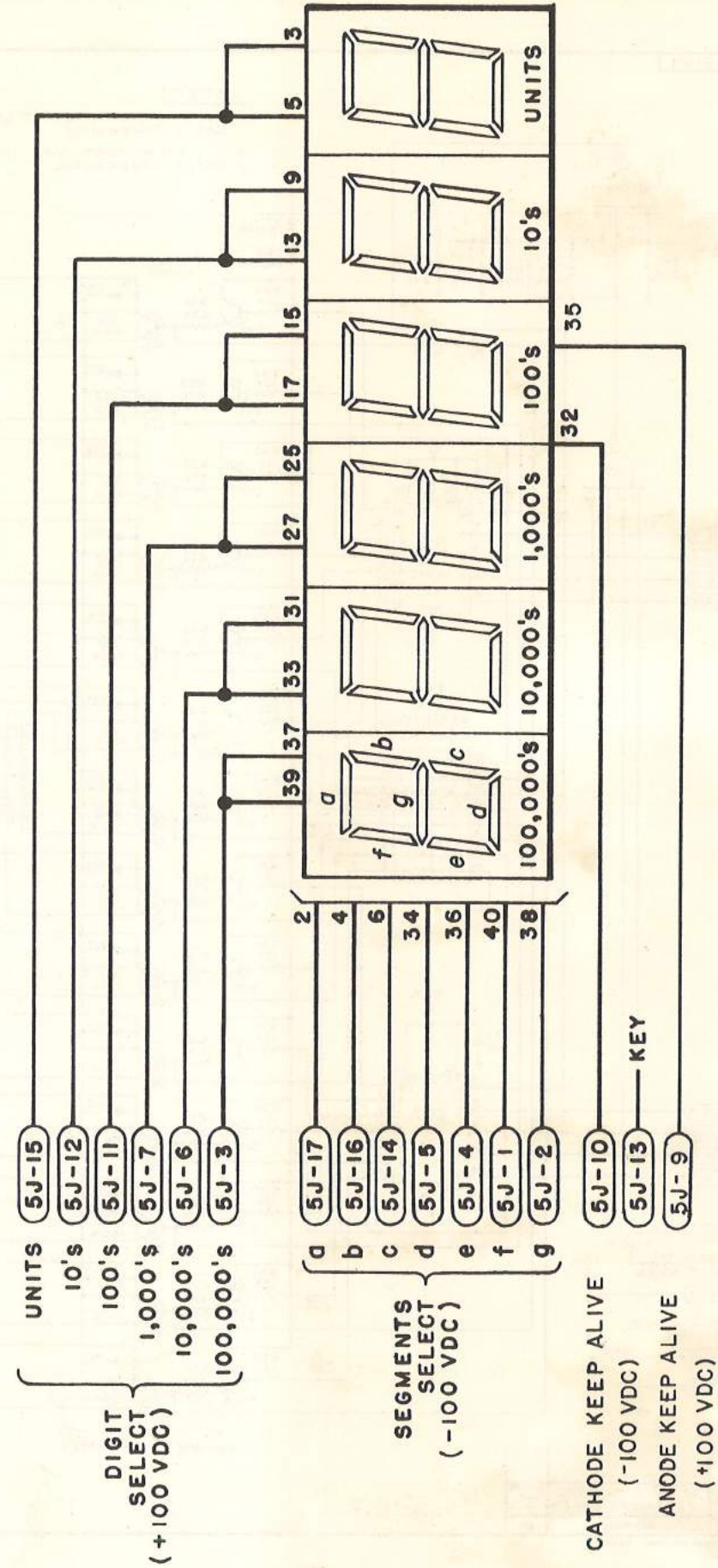
REVISION LETTER	REVISION
	REVISED AND REDRAWN R. GAY 7-19-78

BILL OF MATERIAL				
ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'd. NO.
1	1B-2001-132		BARE P.C. BOARD	1
2	5A-8966	11	6 DIGIT DISPLAY	1
3	23A-6934		DISPLAY MTC. ADHESIVE FOAM	1

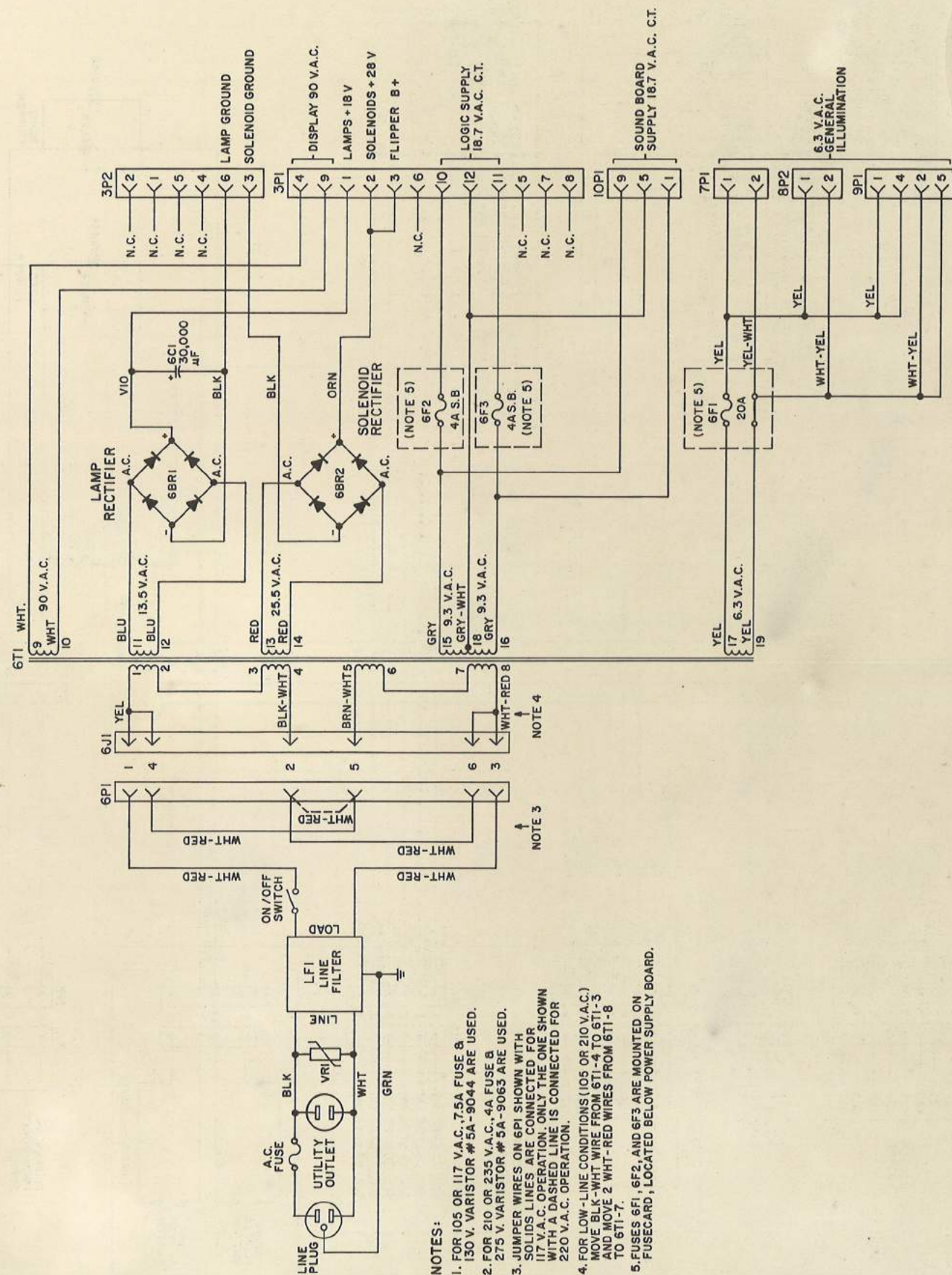


WILLIAMS ELECTRONICS, INC. SUBSIDIARY OF XCOR CORPORATION 3401 N. CALIFORNIA CHICAGO, ILL. 60618 CORNELIA 7-2240				
PART NAME	DATE	APP'D.	SCALE	PART NO.
SLAVE DISPLAY BOARD ASSEMBLY	R. Gay		2:1	C-8019
				DOCUMENT #1

SLAVE DISPLAY BOARD

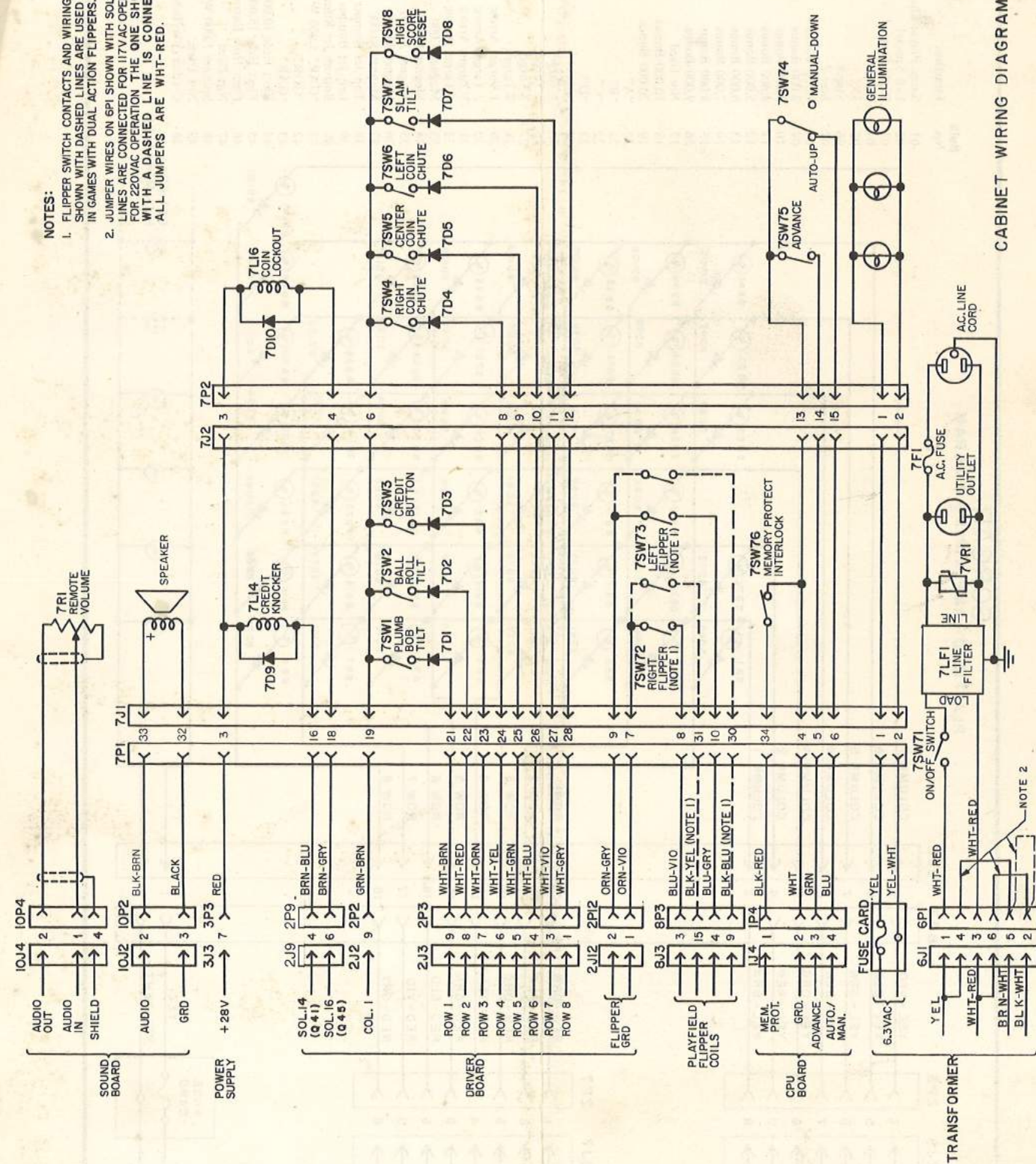


POWER WIRING



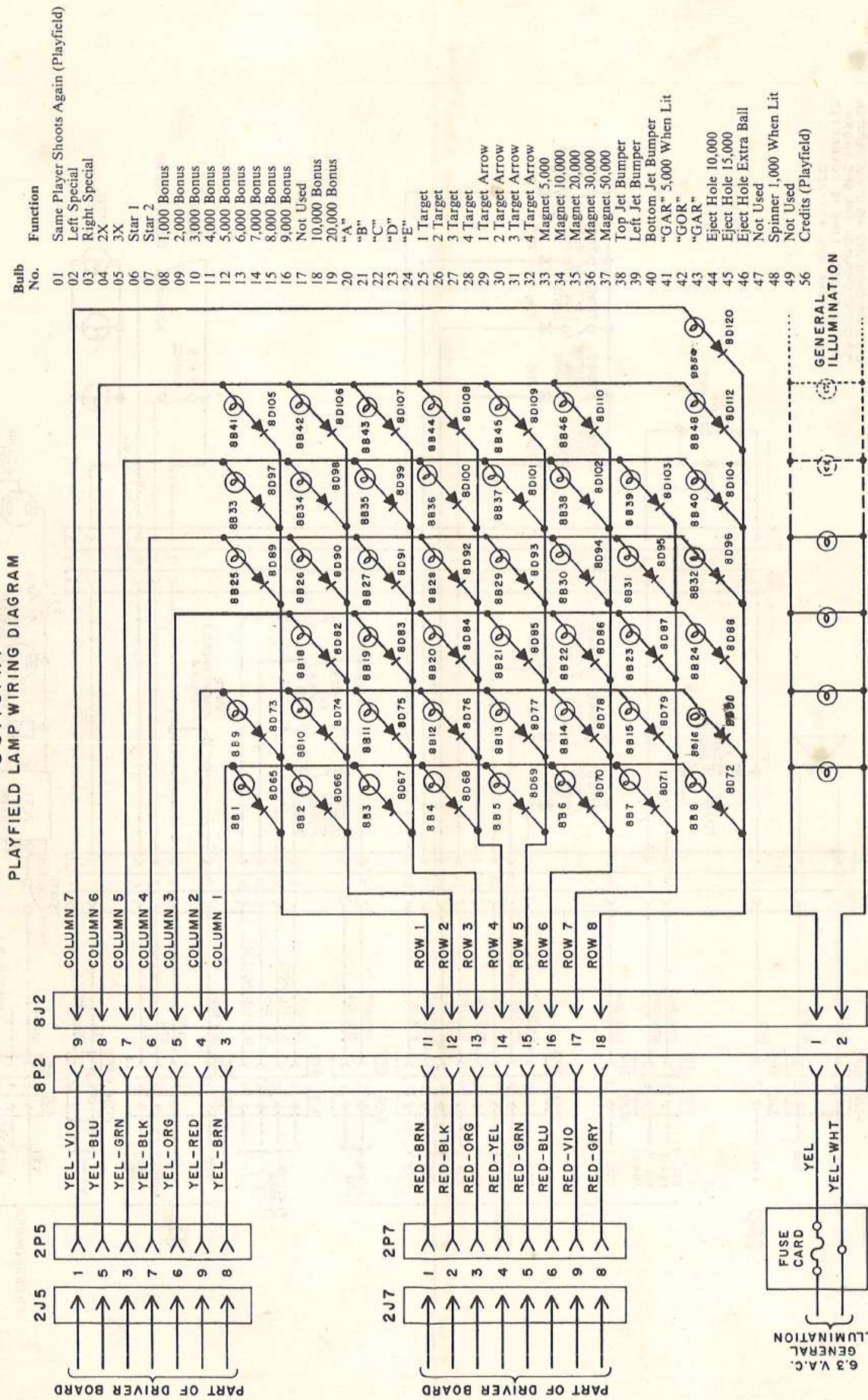
NOTES:

1. FLIPPER SWITCH CONTACTS AND WIRING SHOWN WITH DASHED LINES ARE USED ONLY IN GAMES WITH DUAL ACTION FLIPPERS.
2. JUMPER WIRES ON GPI SHOWN WITH SOLID LINES ARE CONNECTED FOR 117VAC OPERATION, FOR 220VAC OPERATION THE ONE SHOWN WITH A DASHED LINE IS CONNECTED. ALL JUMPERS ARE WHT-RED.



CABINET WIRING DIAGRAM

GORGAR PLAYFIELD LAMP WIRING DIAGRAM



GORGAR PLAYFIELD SOLENOIDS WIRING DIAGRAM

